

Review

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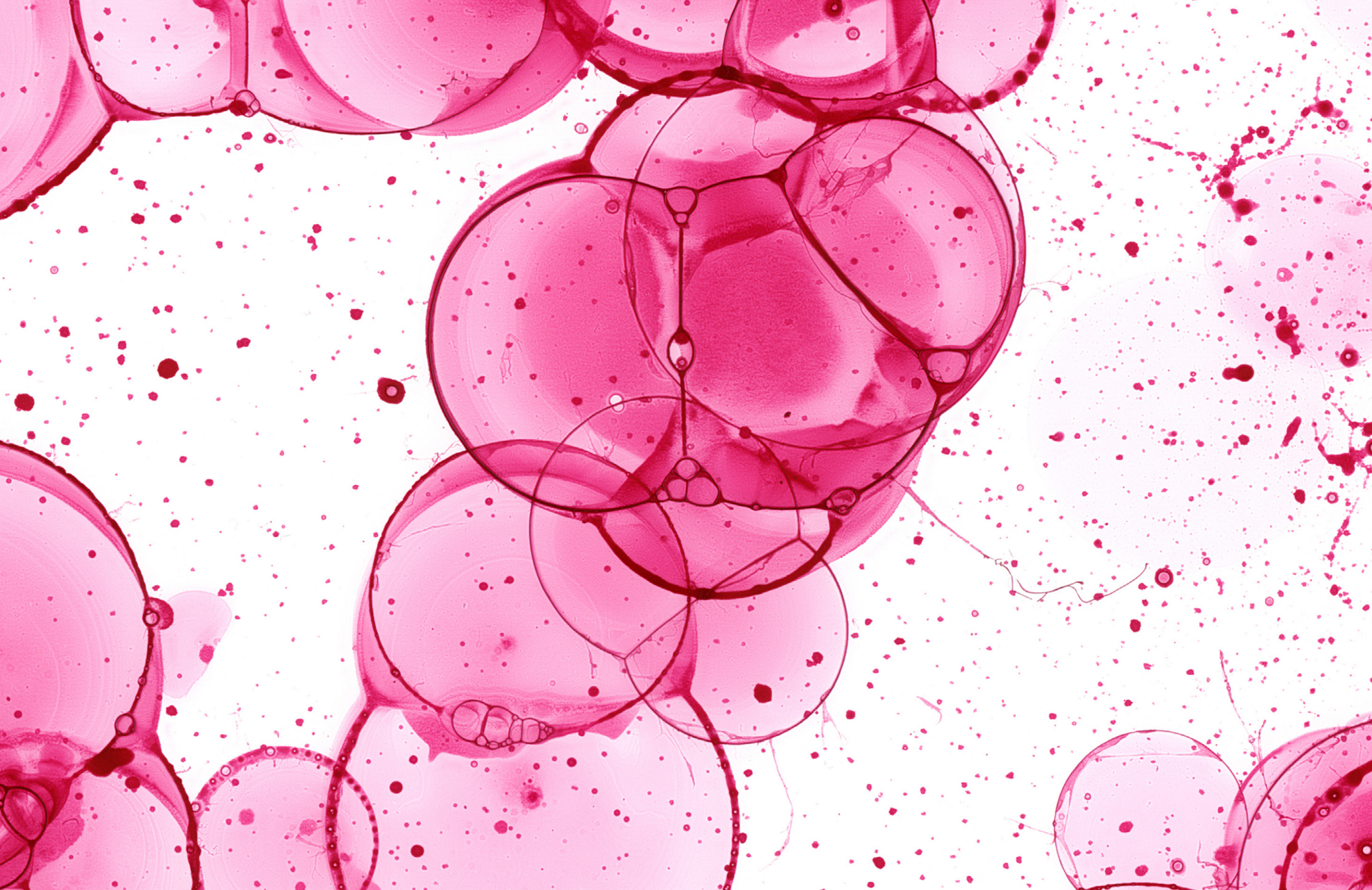
COGNITIVE
AUTOMATION AS
AN IOT ENABLER

FIVE TECH TRENDS
DRIVING INNOVATION

MICROWAVE
BACKHAUL
EVOLUTION



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08 EVOLVING LTE TO FIT THE 5G FUTURE

LTE is one of the most successful mobile communication technologies in the world, and is set to play a major role in mobile communications for many years to come. The process of making it 5G-ready involves a variety of enhancements and new features in Rel-14 and Rel-15, including improved user data rates and system capacity with FD-MIMO, improved support for unlicensed operations, and latency reduction in both control and user planes.



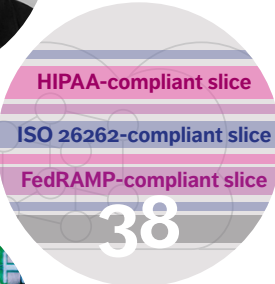
24 MICROWAVE BACKHAUL EVOLUTION: REACHING BEYOND 100 GHZ

No matter how efficiently we use it, existing spectrum will not be sufficient to meet future requirements on network performance. Both radio access and backhaul will need more spectrum in the mid to long term. In light of this, work has started on the use of frequencies beyond 100GHz, enabled largely by advances in high-frequency semiconductor technology.



38 SECURING THE CLOUD WITH COMPLIANCE AUDITING

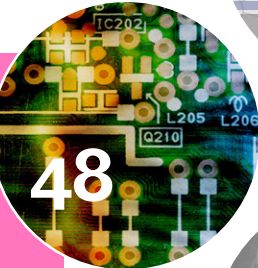
To gain and retain user trust, cloud providers must be able to deploy tenants' applications, store their data securely and ensure compliance with multiple regulations and standards. Moving toward a continuous automated compliance verification model that provides tenants with complete compliance visibility is the key to successfully managing security risks in the cloud.



48 FEATURE ARTICLE

Technology trends driving innovation – Five to watch

The five trends presented here are based on our CTO's understanding of the ongoing transformation of the industry, including rapid digitalization, mobilization and continuous technology evolution, and how this transformation will affect the future development of network platforms.



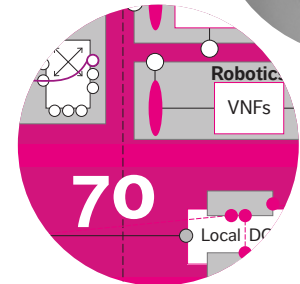
60 TACKLING IOT COMPLEXITY WITH MACHINE INTELLIGENCE

IoT-based systems require a high level of decision making automation both in terms of infrastructure management and within the logic of the IoT applications themselves. Our cognitive automation framework speeds up the development and deployment of intelligent decision support systems (DSSs) by reusing as much knowledge as possible, including domain models, behaviors and reasoning mechanisms.



70 DEVOPS: FUELING THE EVOLUTION TOWARD 5G NETWORKS

Ericsson has worked closely with open source communities such as OPNFV and academic partners to define DevOps as it applies to next-generation telecom networks, identifying the specific steps of the DevOps cycle that are most relevant for 5G infrastructure. This work has resulted in the creation of a DevOps reference pipeline for a 5G business slice, as well as processes and advanced features supporting dynamically software-defined network functions and infrastructure.



NETWORK PLATFORMS:

A CORNERSTONE OF THE EMERGING DIGITAL ECONOMY

■ **AS ERICSSON'S NEW CTO**, I am excited to take over the role of publisher of Ericsson Technology Review and continue the excellent work of my predecessors since the first article was published 93 years ago. I want to take this opportunity to welcome both new and longstanding readers in joining us to gain more technology insights from Ericsson's Research and Development units.

Rapid digitalization, mobilization and continuous technology evolution are all having profound effects on the ongoing development of network platforms, which are a cornerstone of the emerging digital economy. Within, beneath and between these megatrends are a variety of technology trends that we must understand and leverage as we continuously move forward in our work to create top-notch next generation solutions. This year's technology trends article outlines what I consider to be the 'five to watch' in our industry in the years ahead, namely: an adaptable technology base, *the dawn of true machine intelligence, end-to-end security and identity for IoT, an extended-distributed IoT platform, and overlaying reality with knowledge.

Two of the other articles in this issue are closely related to the tech trends article. The first explores how machine intelligence can be used to enhance human decision making ability in the form of decision support systems (DSSs) that automate the management of IoT-based systems. The second touches upon the topic of end-to-end security, looking at how the particular challenges of security compliance in the cloud can be overcome as effectively and cost-efficiently as possible.

This issue also contains three other interesting articles that shed light on important topics such as the evolution of LTE to fit the 5G future; an overview of the latest

“ RAPID DIGITALIZATION, MOBILIZATION AND CONTINUOUS TECHNOLOGY EVOLUTION ARE ALL HAVING PROFOUND EFFECTS ON THE ONGOING DEVELOPMENT OF NETWORK PLATFORMS. ”

developments in microwave backhaul; and how DevOps can be used to satisfy demands for faster turnaround in feature development.

LTE is the most successful mobile communication technology in the world and it is sure to play a major role in mobile communications for many years to come. The process of making it 5G-ready involves a variety of enhancements and new features in Rel-14 and Rel-15. The most significant ones are enhancements to user data rates and system capacity with FD-MIMO, improved support for unlicensed operations, and latency reduction in both control and user planes. These enhancements will allow an operator to move the existing LTE deployments to be a part of the overall 5G solution, as a complement to the deployments of New Radio (NR).

Microwave backhaul technology has been used widely over the years and currently connects a large number of network nodes and base stations, ranging from dense city sites to remote rural sites. It is a technology that is worth paying attention to because it plays a significant role in providing reliable mobile network performance and it is well prepared to support both the evolution of LTE and the introduction of 5G. Efforts are now underway to enable microwave backhaul beyond 100GHz, capitalizing on the rapid evolution of high-frequency semiconductor technologies that support devices operating beyond 100GHz.

As part of the transition to 5G, equipment vendors and telecom operators alike are looking to DevOps as a tool to improve their competitiveness. With DevOps it is possible to reduce the turnaround time for feature delivery cycles and boosting feature hit rates through feedback loops. DevOps also helps companies to strike the right

balance between change and stability, by extending the agile software development culture to deployment and operations. The DevOps article in this issue presents the outcome of our efforts to define DevOps for next-generation telecom networks by scaling it in the OPNFV project and working with academic partners.

I hope you find the contents of the magazine as engaging and thought-provoking as I do. All of the articles included here are also available online at www.ericsson.com/ericsson-technology-review



Erik Ekudden

ERIK EKUDDEN

GROUP CTO AND HEAD OF TECHNOLOGY AND ARCHITECTURE

Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion on the potential, practicalities, and benefits of a wide range of technical developments, and help provide an insight into what the future has to offer.

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FEATURE ARTICLE

Technology trends driving innovation
– Five to watch by Erik Ekudden

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Evolving LTE

TO FIT THE 5G future

With 5G research progressing at a rapid pace, the standardization process has started in 3GPP. As the most prevalent mobile broadband communication technology worldwide, LTE constitutes an essential piece of the 5G puzzle. As such, its upcoming releases (Rel-14 and Rel-15) are intended to meet as many 5G requirements as possible and address the relevant use cases expected in the 5G era.

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HIEU DO**
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Since its first commercial deployment by TeliaSonera in December 2009 [1], LTE has become one of the most successful mobile communication technologies worldwide. Currently, there are 537 commercial LTE networks deployed in 170 countries with 1.7 billion subscribers – a number that is expected to rise to a staggering 4.6 billion by 2022 [2].

■ In the seven years that have passed since the launch of LTE, major advances have been made in terms of both performance and versatility. For example, LTE Rel-8 introduced a 20MHz bandwidth with peak downlink (DL) data rates of 300Mbps and uplink (UL) data rates of 75Mbps [3]. Minor expansions were made for Rel-9, such as multicast/broadcast services, location-based services and dual layer beamforming. LTE Rel-10, also known as LTE-Advanced, introduced several new features such as carrier aggregation (CA) to provide up to 100MHz bandwidth as

well as enhanced support for multi-antennas, heterogeneous deployments and relaying [4]. These features enabled peak data rates in excess of 1Gbps in DL and 500Mbps in UL.

Rel-11 and Rel-12 included enhancements such as the support of machine type communications (MTC), dual connectivity (DC), LTE-WLAN radio interworking, and national security and public safety (NSPS) services including direct device-to-device (D2D) communication [5]. Further advances were made in Rel-13, including spectral efficiency enhancements via Full Dimension multiple-input, multiple-output (FD-MIMO), support for utilizing unlicensed spectrum via Licensed Assisted Access (LAA) and LTE-WLAN aggregation, extended support for MTC through Narrowband Internet of Things (NB-IoT) and enhanced MTC (eMTC), enhanced CA (up to 32 carriers), indoor positioning enhancements, and single-cell-point-to-multipoint (SC-PTM) for broadcast/multicast services [6].

Since October 2015, 3GPP has used the term LTE-Advanced Pro for Rel-13 and onwards, signifying that LTE has reached a maturity level that not only addresses enhanced functionality/efficiency but also the support of new use cases.

●● **LTE HAS REACHED A MATURITY LEVEL THAT NOT ONLY ADDRESSES ENHANCED FUNCTIONALITY/EFFICIENCY BUT ALSO THE SUPPORT OF NEW USE CASES** ●●

Why 5G?

Global mobile data traffic is expected to grow at a compound annual rate of 45 percent in the coming years, which represents a tenfold increase between 2016 and 2022 [2]. This increase is driven largely by the massive adoption of mobile video streaming. On top of that, the IoT is shifting from vision to reality, and of the 29 billion connected devices it is expected to include by 2022, 18 billion will be IoT (or machine-to-machine) devices [2]. Future 5G networks will need to support these challenging new use cases in a cost and energy efficient manner.

Abbreviations

AS – access stratum | **BS** – base station | **CA** – carrier aggregation | **CN** – core network | **CP** – control plane | **CSI** – channel state information | **CSI-RS** – CSI reference signal | **D2D** – device-to-device | **DC** – dual connectivity | **DL** – downlink | **DoNAS** – data over non-access stratum | **DSRC** – dedicated short range communications | **eMBB** – enhanced mobile broadband | **eMTC** – enhanced MTC | **eNB** – evolved node B | **FD-MIMO** – Full Dimension MIMO | **HARQ** – hybrid automatic repeat request | **IoT** – Internet of Things | **ITS** – intelligent transportation system | **ITU** – International Telecommunication Union | **LAA** – Licensed Assisted Access | **MBMS** – Multimedia Broadcast/Multicast Service | **MCL** – maximum coupling loss | **MIMO** – multiple-input, multiple-output | **mMTC** – massive machine type communications | **mm-wave** – millimeter wave | **MTC** – machine type communications | **MU-MIMO** – multi-user MIMO | **NAS** – non-access stratum | **NB-IoT** – Narrowband Internet of Things | **NR** – New Radio | **PCell** – primary cell | **RRC** – Radio Resource Control | **RS** – reference signal | **RTT** – round-trip time | **SCell** – secondary cell | **SL** – sidelink | **SR** – scheduling request | **TTI** – transmission time interval | **UL** – uplink | **UP** – user plane | **URLLC** – ultra-reliable low latency communications | **V2I** – vehicle-to-infrastructure | **V2N** – vehicle-to-network | **V2P** – vehicle-to-pedestrian | **V2V** – vehicle-to-vehicle | **V2X** – vehicle-to-everything | **3GPP** – 3rd generation partnership project

THE PROCESS OF MAKING LTE 5G-READY INVOLVES A VARIETY OF ENHANCEMENTS AND NEW FEATURES IN REL-14 AND REL-15

Although the requirements for 5G capabilities are still being finalized both in the ITU [7] and 3GPP [8], there is a preliminary agreement regarding the three main use cases the technology must support. As illustrated in **Figure 1**, they are: enhanced mobile broadband (eMBB), ultra-reliable low latency communications (URLLC) and massive machine type communications (mMTC). eMBB refers to the extended support of conventional MBB through improved peak/average/cell-edge data rates, capacity and coverage. URLLC is a requirement for emerging critical applications such as industrial internet, smart grids, infrastructure protection, remote surgery and intelligent transportation systems (ITSs). Last but certainly not least, mMTC is necessary to support the envisioned 5G IoT scenario with tens of billions of connected devices and sensors.

There are two tracks that make up the 5G radio access roadmap in 3GPP, as illustrated in **Figure 2**. One is based on the evolution of LTE and the other on New Radio (NR) access. In the LTE-5G track, enhancements will continue to enable it to support as many 5G requirements and use cases as possible. Unlike the LTE-5G track, the NR-5G track is free from backward compatibility requirements and thereby able to introduce more fundamental changes, such as targeting spectrum at high (mm-wave) frequencies. However, NR is being designed in a scalable manner so it could eventually be migrated to frequencies that are currently served by LTE.

While the prospects for the NR-5G track are exciting, the operators that have already made significant investments in LTE do not need to be concerned – a transition from LTE to 5G through 5G plug-ins is the most logical course of action. Both the expectations for LTE Rel-14 [9] – which is scheduled for completion in March 2017 – and the strong ambitions for LTE Rel-15 indicate that the development plans for the LTE-5G track are solid.

The process of making LTE 5G-ready involves a variety of enhancements and new features in Rel-14 and Rel-15. The most significant ones are enhancements to user data rates and system capacity with FD-MIMO, improved support for unlicensed operations, and latency reduction in both control and user planes (UPs). The enhancements in Rel-14 and Rel-15 also aim to provide better support for use cases such as massive MTC, critical communications and ITS.

User data rate and system capacity enhancements

FD-MIMO and unlicensed operations are the two main features in the upcoming releases of LTE that are intended to bring about improved user data rates and system capacity that meet 5G standards.

FD-MIMO

The MIMO enhancement in 3GPP makes it possible to dynamically adapt transmission both vertically and horizontally by utilizing a steerable two-dimensional antenna array. The concept of FD-MIMO in future LTE releases builds on the channel state information (CSI) feedback mechanisms introduced in LTE Rel-13, in which precoding matrix codebooks support two-dimensional port layouts with up to 16 antenna ports. Non-precoded CSI reference signals (CSI-RSs) are transmitted from each antenna and broadcast in the cell, and the precoder is derived by the terminal. LTE Rel-13 also introduced another CSI feedback type with terminal-specific, beamformed CSI-RS, in the same fashion as physical downlink shared channel (PDSCH).

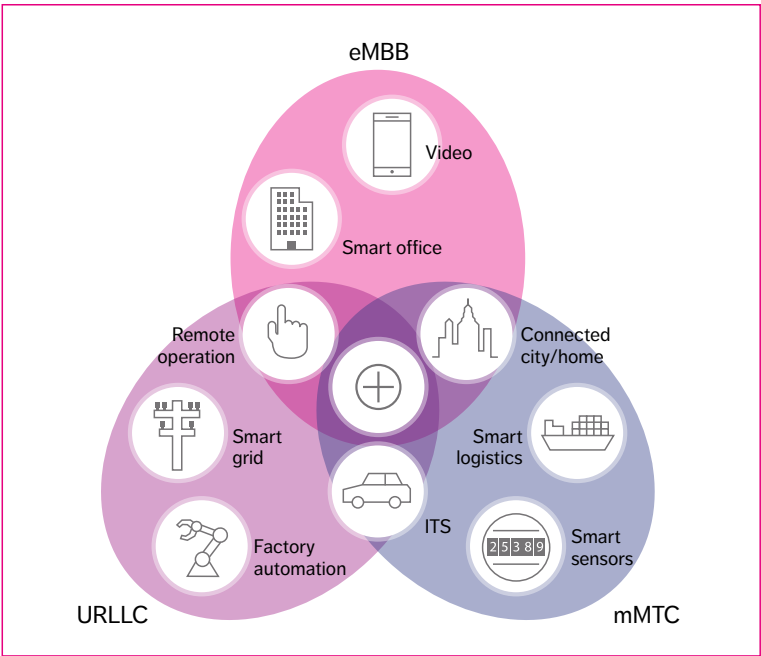


Figure 1
The three main 5G use cases and examples of associated applications

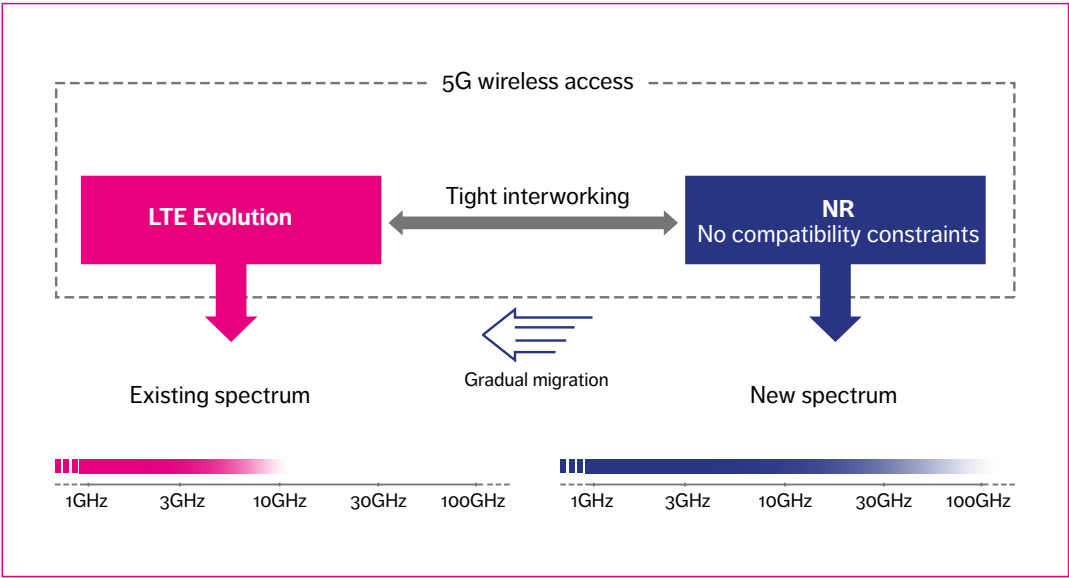


Figure 2
5G radio access roadmap

In this case, the beamforming direction for each terminal is decided by the base station rather than being derived from terminal feedback.

To enhance both non-precoded and beamformed CSI-RS operation, Rel-14 will introduce several new features, including hybrid non-precoded/beamformed CSI mode with optimized feedback; aperiodic triggering of CSI-RS measurements; support for up to 32 antenna ports; spatially rich, advanced CSI feedback; and a semi-open-loop transmission scheme.

Hybrid non-precoded and beamformed CSI mode with optimized feedback will make it possible to intermittently transmit an initial, non-precoded CSI-RS. The terminals can then respond with a desired direction for a second, more frequent, beamformed CSI-RS.

Aperiodic triggering of CSI-RS measurements facilitates CSI-RS resource pooling, enabling the efficient use of measurement resources and the reduction of CSI-RS overhead. As a result, more terminals in the cell will have access to beamformed CSI-RS operation.

Support for 32 antenna ports makes it possible to use feedback-based operation with massive antenna setups, which increases the gains from multi-user MIMO (MU-MIMO).

Spatially rich, advanced CSI feedback will include information about multiple channel propagation paths, so that interference between co-scheduled terminals can be avoided or suppressed. Performance is then comparable to reciprocity-based massive MU-MIMO systems.

The semi open-loop transmission scheme combines full-dimension beamforming and transmit diversity, targeting high-speed terminals where a beam direction is known but short-term CSI changes too quickly.

The anticipated improvement in system capacity and user throughput with Rel-14 FD-MIMO is illustrated in **Figure 3** – a 3GPP 3D urban micro scenario featuring 8x4 dual polarized array and non-full-buffer traffic. Performance on the cell edge increases roughly 2.5 times with advanced CSI feedback and support for 32 antenna ports.

SEVERAL SOLUTIONS HAVE BEEN INCORPORATED INTO 3GPP TO ACHIEVE COEXISTENCE WITH OTHER TECHNOLOGIES – SUCH AS WLAN – THAT OPERATE IN THE SAME BAND AS LAA

LTE operations in unlicensed spectrum

To address ever increasing traffic demands, many network operators are considering complementary use of unlicensed spectrum. LAA was introduced in LTE Rel-13 for DL operation, and it is being enhanced in Rel-14 to support UL. LAA uses CA to combine a licensed band primary cell (PCell) with unlicensed band secondary cells (SCells). The SCells usually have restricted transmission power, however, which results in coverage areas that are smaller than those that PCells are able to provide. In this arrangement, a PCell provides reliable coverage for control messages and high-priority traffic, while the SCells provide a large amount of spectrum and high data rates when available. **Figure 4** shows how LAA offers a combination of the main benefits provided by both licensed and unlicensed spectrum.

Several solutions have been incorporated into 3GPP to achieve coexistence with other technologies – such as WLAN – that operate in the same band as LAA. These include dynamic carrier measurement/selection, Listen-Before-Talk protocol, and discontinuous transmission with limited maximum duration. Smart and adaptive traffic management between licensed and unlicensed carriers – and between unlicensed carriers – could also further enhance coexistence. **Figure 5** shows the network capacity in an LAA outdoor coexistence scenario where each of

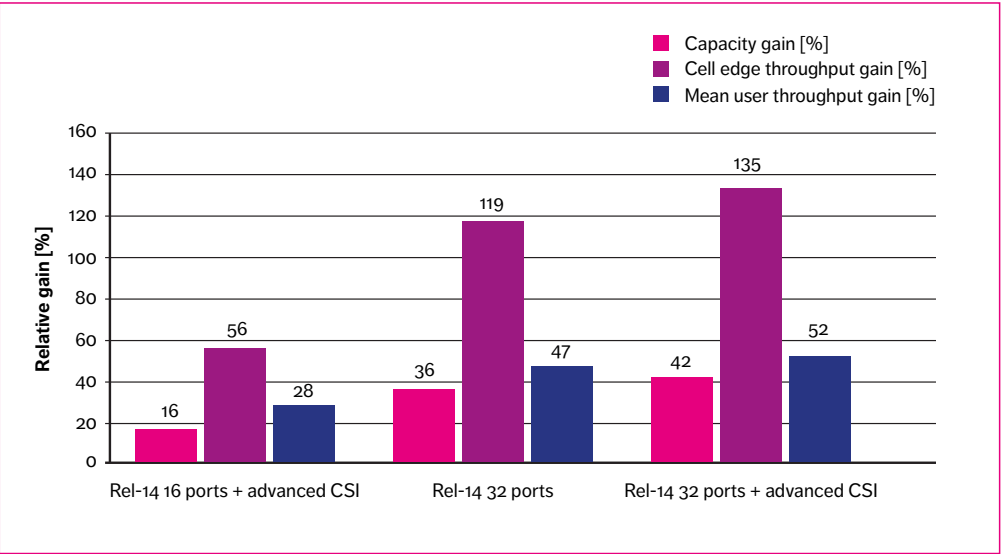


Figure 3 Performance of Rel-14 FD-MIMO over a 16 port Rel-13 baseline (without advanced CSI) at high system load

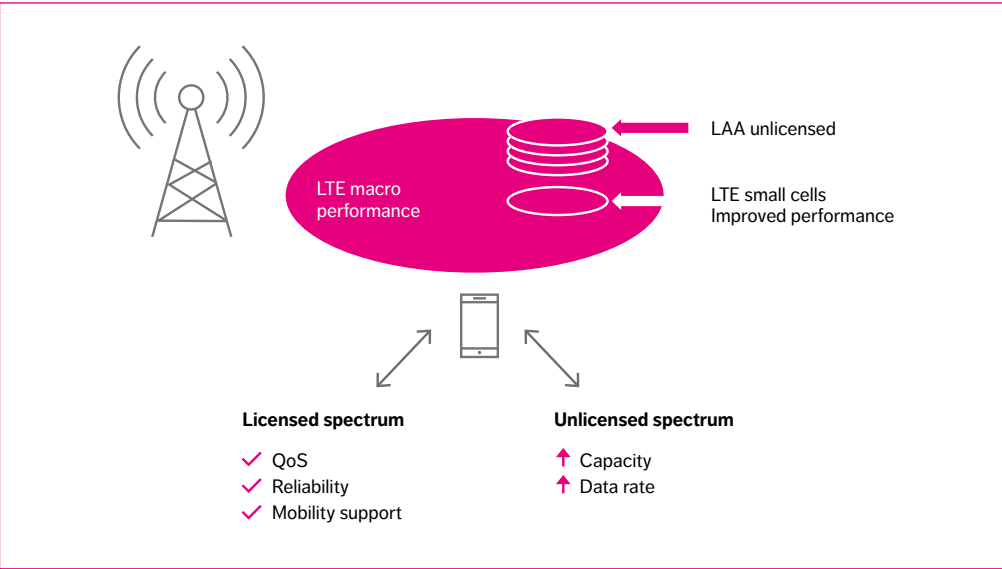


Figure 4 Illustration of LAA

●● LATENCY REDUCTION NOT ONLY CONTRIBUTES TO DATA RATE ENHANCEMENTS BUT ALSO ENABLES NEW USE CASES SUCH AS CRITICAL COMMUNICATION AND ITS ●●

two operators deploy four LAA or four WLAN nodes per hotspot [10]. The LAA cells support substantially higher offloading capacity on the same 20MHz channel compared with the WLAN nodes. This is because the robust LAA physical layer design allows reliable and efficient frequency reuse. In fact, the more efficient LAA network leaves more capacity for the co-channel WLAN.

Further LAA enhancements are expected in LTE Rel-15, most notably UL control information transmission and random access channel support on the unlicensed band SCells. This would make it possible to offload more traffic from the licensed band PCells and allow for further deployment as well as enabling use cases such as fiber connected remote radio heads.

Another potential enhancement in LTE Rel-15 is dual connectivity between licensed band main evolved node B (eNB) and unlicensed band secondary eNB. This would further broaden deployment possibilities by allowing aggregation between network nodes that are not connected via low-latency backhaul. Finally, Rel-15 may enable more deployment options and scenarios, such as standalone and mMTC operations in unlicensed spectrum.

Latency reduction

Another important aspect of LTE enhancement is the implementation of latency reduction techniques for the user and control planes (UPs and CPs). Latency reduction not only contributes

to data rate enhancements but also enables new use cases such as critical communication and ITS.

User plane latency reduction

Implementing fast UL access is the first step toward reducing UP latency. As specified in Rel-14, fast UL access makes it possible to configure a terminal with an uplink grant available in each millisecond, to be used only when there is uplink data to transmit. Using the current scheduling request (SR) based access, the terminal must transmit a request, wait for a grant, and then wait to use the grant. A comparison of fast UL access with SR access is illustrated in the a and b tracks of *Figure 6*. The pre-configured grant in fast UL access minimizes the waiting time, which reduces the average radio access delay for uplink data by more than half.

The other latency reduction step consists of two enhancements that are both targeted for specification in Rel-15. The first is reduced processing time: making the terminal respond to downlink data and uplink grants in three milliseconds instead of four. The second is the introduction of shorter transmission time intervals (TTIs): speeding up the whole chain of waiting for a transmit opportunity, scheduling and preparing for a transmission, transmitting the data, and ultimately processing the received data and sending feedback.

With a short TTI, as illustrated in the c track of *Figure 6*, transmissions can be made with a shorter duration (as little as one-seventh of the length of a normal LTE TTI). Each of these short transmissions can be scheduled separately with a new DL in-band control channel, with feedback sent in a new UL control channel. The scheduling and feedback are sent in adjacent subframes for the shortest transmission time, resulting in a total radio access one-way transmission delay of about 0.5ms, including data processing time.

Figure 7 illustrates the gains in round-trip time (RTT) made by employing short TTI and fast UL access. From simulations, improvements have also been observed in the throughput for File Transfer

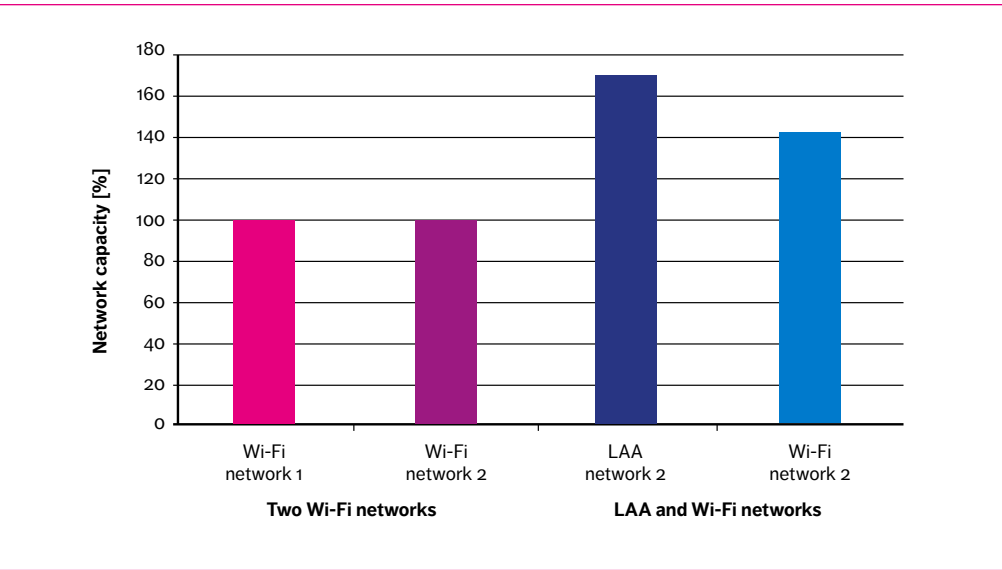


Figure 5
LAA-WLAN outdoor coexistence (40MHz shared carriers, both networks operating at 5GHz)

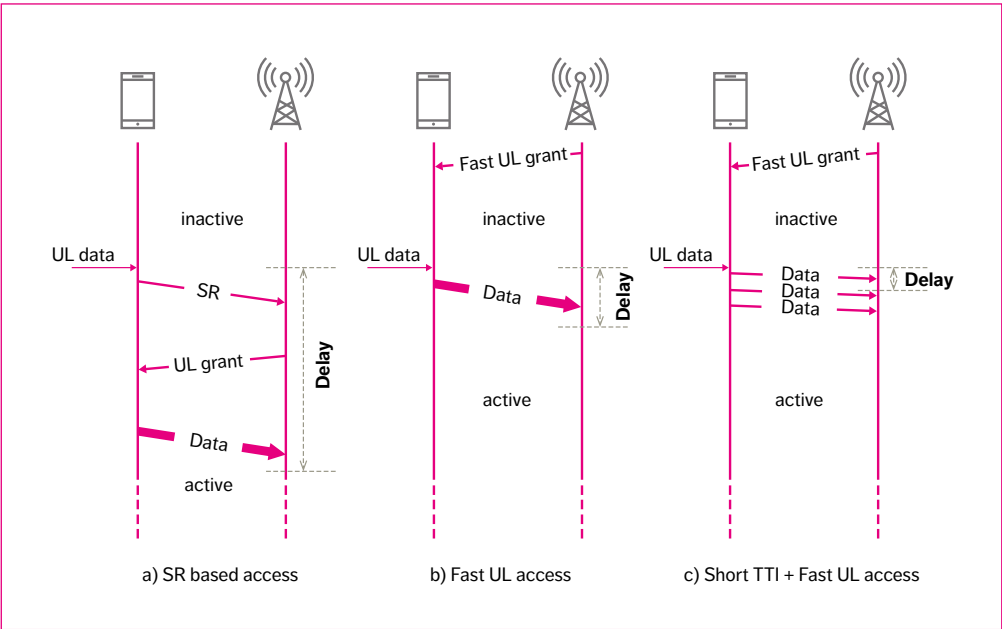


Figure 6
SR access (a), fast UL access (b), and short TTI in conjunction with fast UL access (c)

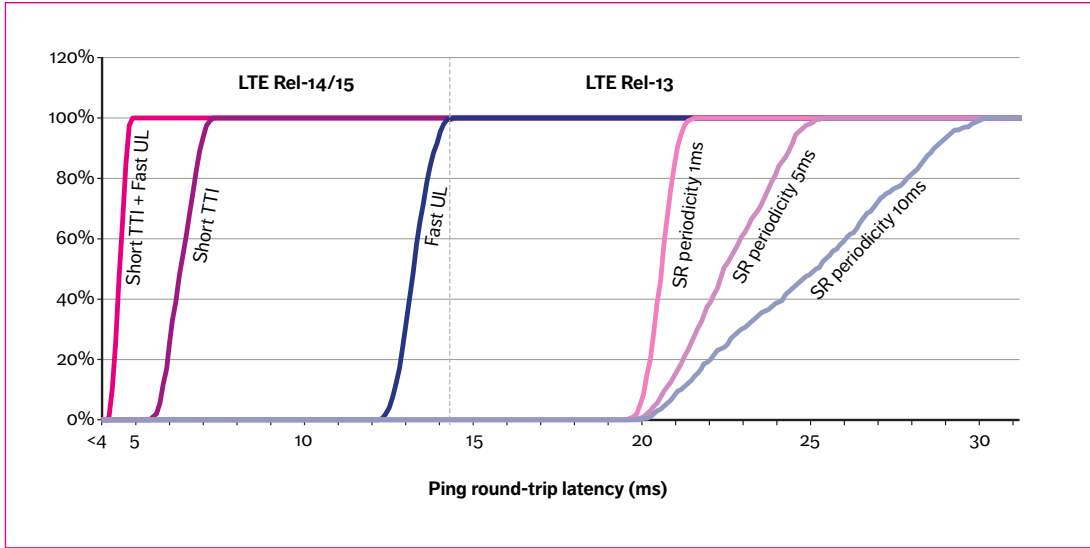


Figure 7 Impact of short TTI and fast UL access on RTT

Protocol (FTP) download by up to 70 percent: an effect caused by a faster TCP bitrate ramp-up thanks to the shorter RTT of data and response.

Signaling reduction

LTE state transitions involve significant signaling: going from RRC_IDLE to RRC_CONNECTED comprises 9 transmissions over the air interface. Two options for signaling reduction were introduced in Rel-13: RRC connection suspend/resume for use with UP based data transfer over data radio bearers (DRBs) and data over non-access stratum (DoNAS) for CP-based data transfer over the signaling radio bearer (SRB).

The suspend/resume feature allows the data connection to be suspended temporarily and the context to be stored in the RAN and core network (CN) during RRC_IDLE. At the next transition to RRC_CONNECTED, the connection is resumed with the stored context, significantly reducing

the signaling to four or five transmissions. The DoNAS feature achieves a similar reduction of signaling by omitting access stratum (AS) security and by transferring data over the CP instead of establishing traditional UP radio bearers.

To accommodate the ever increasing number of devices, small and/or infrequent data volumes and stricter delay requirements, Rel-14 and Rel-15 aim for further reduction of signaling between terminals and network nodes (RAN and CN).

In Rel-14, the suspend/resume feature is being improved by reducing the signaling between the base station (BS) and the CN. In Rel-13, the BS-CN connection was released together with the air interface connection. In Rel-14, the BS-CN connection can be kept when the BS-terminal connection is suspended. The RAN takes over the responsibility of paging the terminal upon the arrival of DL data, for example.

Two additional control plane latency reduction

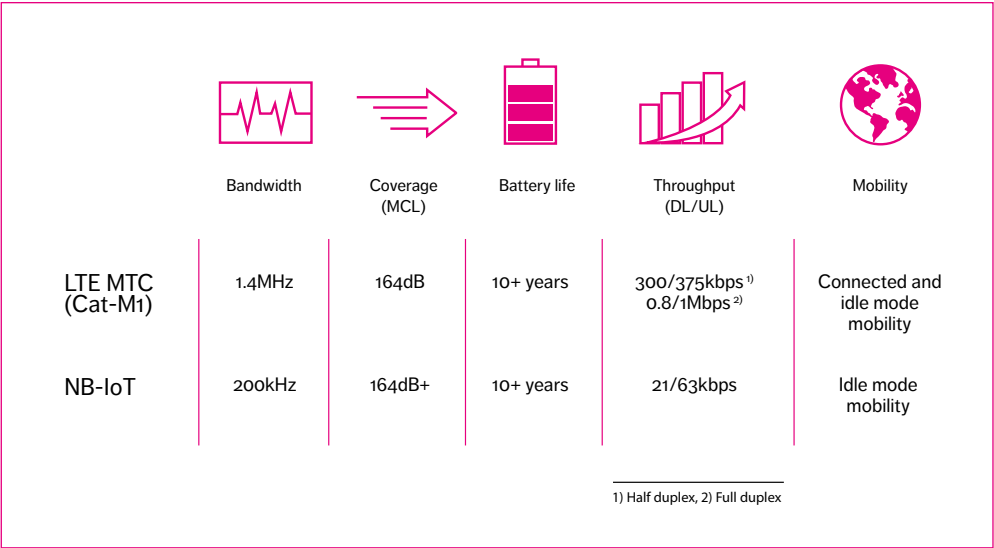


Figure 8 NB-IoT and LTE MTC key performance indicators (Rel-13)

improvements are expected in Rel-14 or Rel-15. The first is an enhancement that would enable earlier data transmission by making it possible to multiplex UP radio bearer data with connection resume signaling. The second is known as release assistance indication, which would allow the terminal to indicate that it has no more UL data and that it does not anticipate DL data, thereby enabling early transition to RRC_IDLE.

New use cases for 5G

A number of improvements in LTE Rel-14 and Rel-15 are designed to provide improved support for use cases such as massive MTC, critical communications and ITS.

Massive machine type communications

LTE MTC and NB-IoT were developed to address mMTC use cases [11]. They offer similar improvements with regard to coverage

enhancement, battery life, signaling efficiency and scalability, but address slightly different demands in terms of flexibility and performance. As shown in Figure 8, LTE MTC is more capable of supporting higher data rates and both intra-RAT and inter-RAT connected mode mobility. With the new LTE MTC Category M1 (Cat-M1) and NB-IoT, which were specified in 3GPP Rel-13, it is anticipated that modem cost can be drastically reduced compared with Rel-8 Cat-1 devices. Cost will vary depending on features, options and implementation. Modem cost reductions are expected to be in the order of 75-80 percent for Cat-M1 [12] and even more for NB-IoT with its further reduced feature set.

LTE Rel-14 aims to further enhance LTE MTC and NB-IoT by improving performance and addressing more use cases. Higher data rates and efficiency will be achieved in Rel-14 by allowing larger chunks of data to be carried in each

transmission and increasing the number of hybrid automatic repeat request (HARQ) processes to enable parallel outstanding transmissions while waiting for feedback. Larger channel bandwidth for LTE MTC (up to 5MHz) enhances support for voice and audio streaming as well as other applications and scenarios. NB-IoT enhancements for random access and paging increase the versatility of non-anchor carriers.

Rel-14 will further enable positioning applications (in which knowledge of device location is critical) by supporting enhanced reference signals that take into account the smaller NB-IoT/LTE MTC bandwidth. Enhancements to connected mode mobility will improve service continuity. Multicast transmission will make the delivery of the same content to multiple devices more efficient, optimizing use cases such as firmware upgrades and synchronous control of things like streetlights, for example. Support for the lower NB-IoT power class of 14dBm will enable the use of smaller batteries and support devices with a small form factor.

Voice coverage for LTE MTC will be improved in Rel-14 by increasing VoLTE coverage for half-duplex FDD/TDD through techniques that reduce DL repetitions, new repetition factors, and adjusted scheduling delays. MTC devices and use cases will also benefit from the signaling reduction enhancements in LTE Rel-14.

mMTC use cases will also benefit from a few other enhancements in LTE Rel-15, including:

- » latency improvements resulting from the multiplexing of user data with connection resume signaling
- » efficiency improvements resulting from enhanced access/load control in idle and connected modes
- » battery life improvements resulting from relaxed DL monitoring requirements in idle mode
- » improved support for additional use cases such as wearables.

Critical communication

Use cases such as power grid surveillance, safety-critical remote control, and critical manufacturing

operations require both low latency and high reliability above the current HARQ level (see Figure 9). In order for LTE to meet these 5G requirements, there is an aim for two improvements to be made for Rel-15: reliable short TTI operation and reliable 1ms operation.

By building on the short TTI and fast UL features, the packet error rate can be reduced to a 10⁻⁵ level through a combination of robust coding of control and data messages, diversity, and automatic repetitions without feedback. Since the processing is kept on a short timescale, the entire chain of transmissions can be delivered within 1ms with the combined reliability of multiple trials. (The target is small cells, such as factories and offices.) In addition, wide-area coverage with relaxed latency but extreme reliability can also be targeted by automatic repetitions of robustly coded 1ms transmissions with enhanced feedback.

Intelligent transportation systems

The use of ICT to enable safer and more efficient transportation systems is known as ITS. 3GPP has been developing a solution for vehicle-to-everything (V2X) communications for Rel-14, addressing the connection between vehicles (vehicle-to-vehicle or V2V), vehicle-to-network (V2N), vehicle-to-infrastructure (V2I), and vehicle-to-pedestrian (V2P), as illustrated in Figure 10.

LTE-based ITS benefits from the coverage of the existing networks and the centralized security. However, new ITS use cases are demanding in terms of latency and system capacity. Therefore, the direct D2D interface, known as sidelink (SL), and the LTE cellular air interface are being enhanced in Rel-14 to support these requirements.

For example, increased pilot symbol density will make it possible to optimize the SL for quickly changing propagation conditions and severe frequency shifts at the receiver due to high relative speed (up to 500km/h) and higher carrier frequency (up to 6GHz).

Improved radio resource management is another important enhancement to support ITS

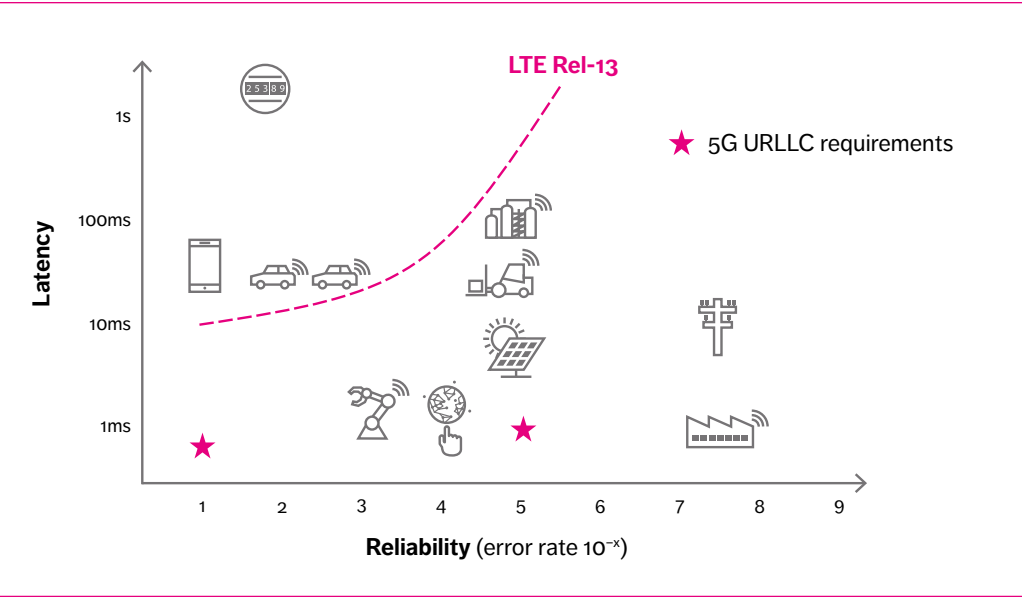


Figure 9
Critical communication use cases and requirements

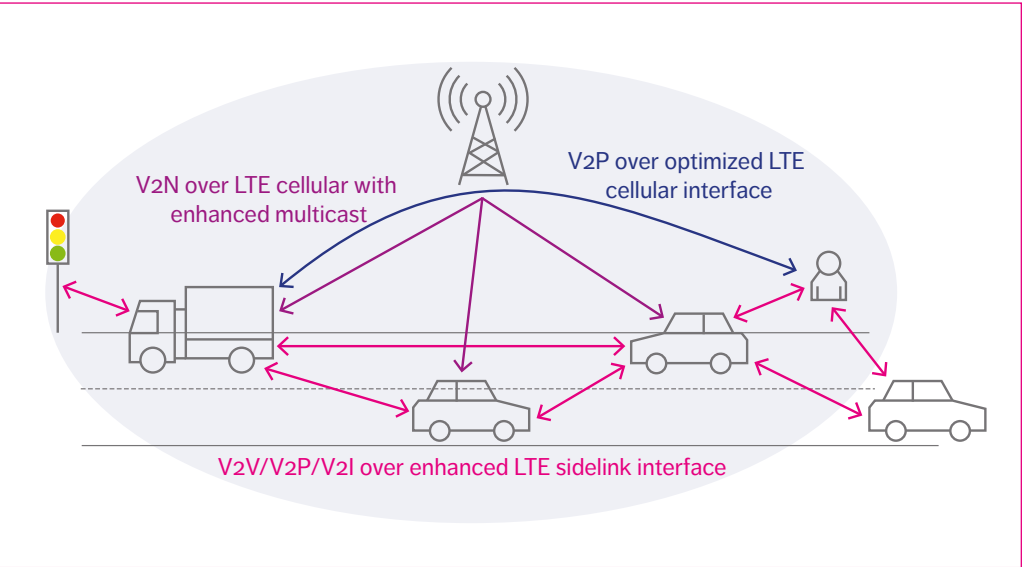


Figure 10
Illustration of different ITS scenarios and interfaces

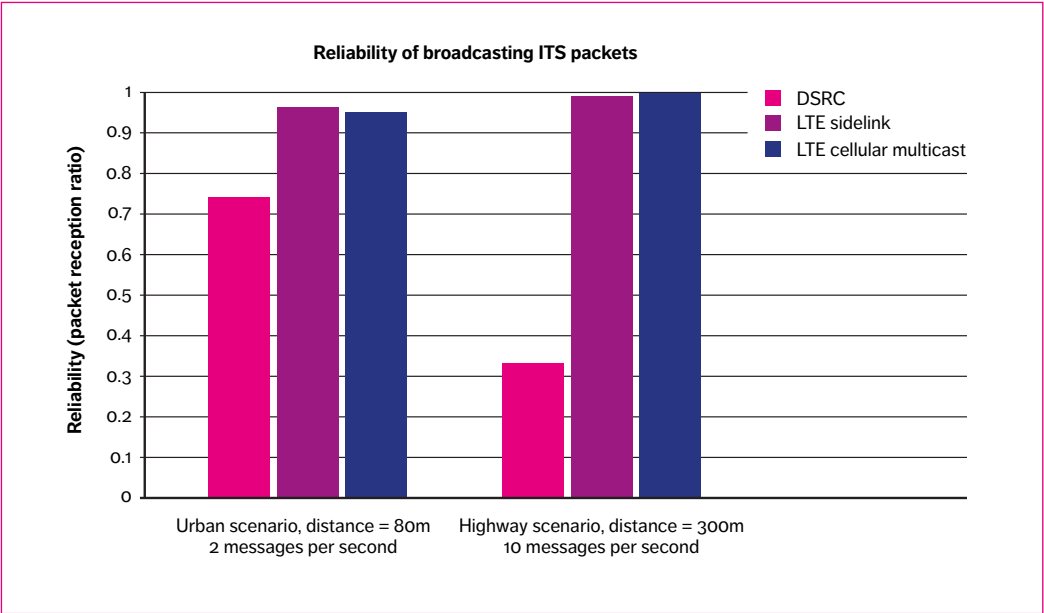


Figure 11 Comparison of different technologies for broadcasting ITS messages

applications. This is based on a sensing-based resource selection protocol, where each device autonomously learns how other devices use the radio resources and predicts their future behavior, taking advantage of the quasi-periodic nature of the ITS messages.

Rel-14 supports the usage of geographical location information to enable centralized resource allocation in the eNB or to autonomously select a resource within a configured radio resource pool. It also supports Multimedia Broadcast/Multicast Service (MBMS) protocols that are optimized for low latency and coverage, and efficient delivery of V2X messages. Finally, the expected enhancements will provide fair and efficient coexistence with

non-3GPP ITS technologies such as dedicated short range communications (DSRC).

Figure 11 shows a numerical comparison of the capability of different technologies for broadcasting V2V messages. In typical scenarios (urban and highway), the solutions based on LTE (SL with centralized resource allocation and cellular multicast) perform significantly better than the one based on DSRC.

Conclusion

LTE is well positioned to deliver on all the most important 5G requirements, including user data rate and system capacity enhancements with FD-MIMO, improved support for unlicensed

operations, and latency reduction in both user plane and signaling. The improvements planned in Rel-14 and Rel-15 will not only ensure that LTE will provide better support for massive MTC and ITS; they will also enable LTE to address new use cases such as critical communications. *

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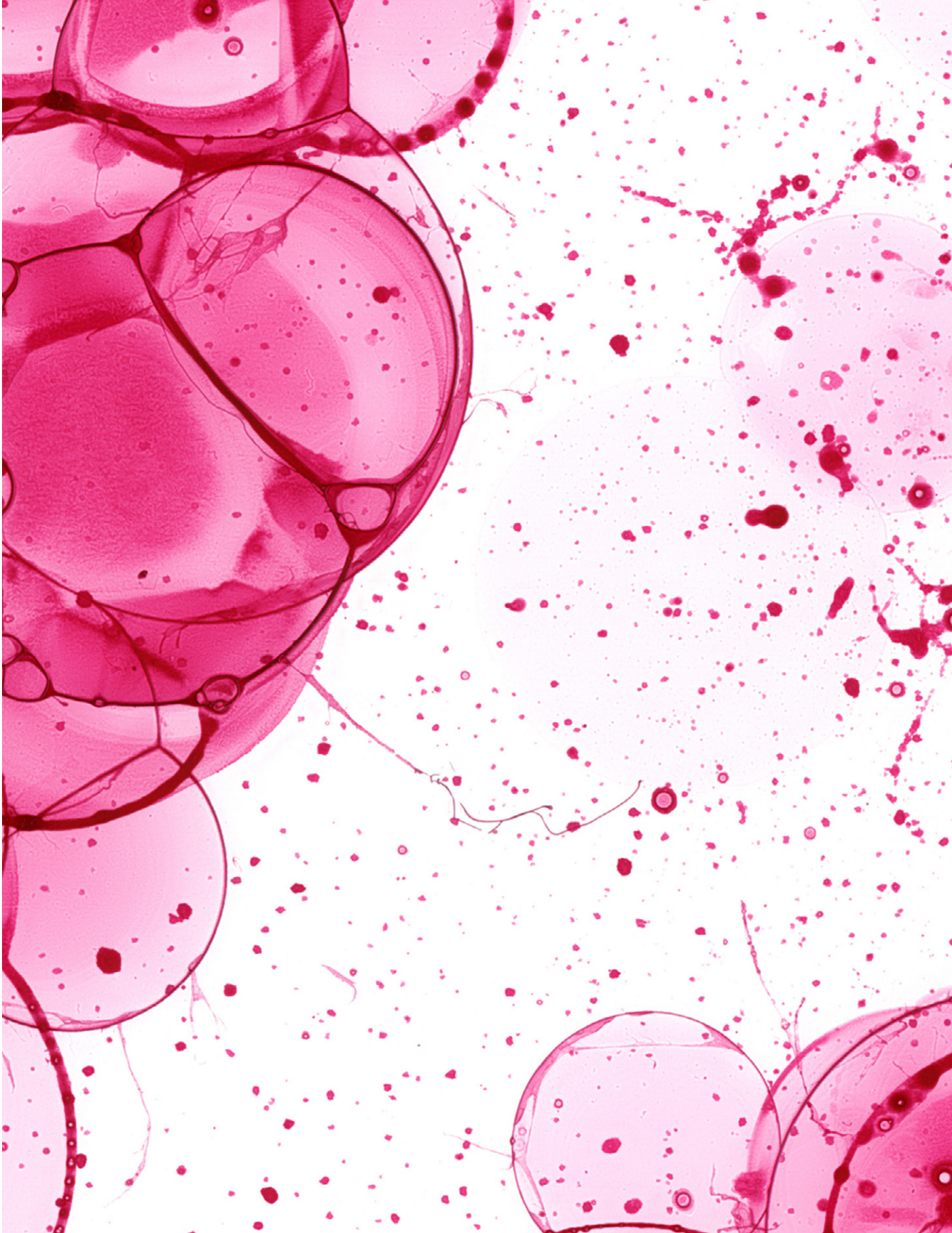
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MICROWAVE

backhaul evolution

– REACHING BEYOND 100GHz

Microwave backhaul technology plays a significant role in providing reliable mobile network performance and is well prepared to support both the evolution of LTE and the introduction of 5G. Work has now started on the longer-term use of frequencies beyond 100GHz, targeting the support of 5G evolution toward 2030.

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Constant pressure to improve performance levels results in a need for more spectrum, and the more efficient use of it – not just for radio access, but for backhaul as well. By continuously pushing technology limits, ever higher frequencies have been brought into use during the last few decades – a trend that will continue in the future.

■ As a finite natural resource, radio spectrum is governed by national, regional and international regulations to ensure that social and economic benefits are maximized. Spectrum is divided into frequency bands that are allocated to different

types of radio services, such as communication, broadcasting and radar, as well as for scientific use [1].

By 2021, 65 percent of the world’s cell sites (excluding those in northeast Asia) will be connected using microwave backhaul technology [2]. The rapidly growing capacity requirements that this entails will create a need for significant performance improvements enabled by technology evolution and more efficient use of existing spectrum [2, 3, 4].

The microwave backhaul industry has started preparing for the next major technology and performance leap to accommodate the market’s expected volume needs for the 2025 to 2030

period. Making such leaps requires many years of research and development and a great deal of work on spectrum regulation, as well as the experience of several technology and product generations to mature performance for large-scale use. The aim is to open up spectrum beyond 100GHz frequencies for up toward 100Gbps capacity to support different applications and use cases with hop distances of up to a few kilometers. In the longer term, it is expected to serve as a high-capacity complement to the use of other frequency bands [2], especially in urban and suburban areas, as shown in *Figure 1*. The smaller physical antenna size at these higher frequencies will be of particular advantage in these locations.

Higher frequencies are more limited in terms of reach and coverage, but they can generally provide wider frequency bands, and as such have higher data-carrying capacities. Driven by growing communication needs, ever higher frequencies have been taken into use since the middle of the last century when the use of frequencies of just a few GHz was the norm for microwave transmission networks. At present, the 70/80GHz band – 71-76GHz paired with 81-86GHz – is rapidly gaining popularity, as it enables capacities in the 1-20Gbps range over a few kilometers [2, 3]. It has taken about 15 years from the initial efforts in this band for large-scale usage to start taking off. Similar efforts are now underway to enable the use of frequencies beyond 100GHz [5, 6] for capacities in the 5-100Gbps range over distances comparable to 70/80GHz today.

●● THE AIM IS TO OPEN UP SPECTRUM BEYOND 100GHz FREQUENCIES FOR UP TOWARD 100GBPS CAPACITY ●●

Microwave backhaul beyond 100GHz

Microwave backhaul or fixed service systems (as they are known in ITU-R terminology) are commonly used in a multitude of frequency bands ranging from 6-86GHz. The range of frequency bands is needed to provide backhaul for diverse types of locations, from sparse rural areas to ultra-dense urban environments, with hop distances ranging from as little as 100m to 100km or more. The use of frequency bands is governed by regulatory recommendations on channel arrangements [7]. Beyond 100GHz, spectrum has been allocated for fixed service systems up to 275GHz [1], but no channel arrangements have been made. However, regulatory studies on channel arrangements are ongoing in Europe [5], with the focus on the 92-114.25GHz and 130-174.8GHz ranges: commonly referred to as the W and D band respectively.

Terms and abbreviations

BER – bit error rate | **BPSK** – binary phase shift keying | **CMOS** – complementary metal-oxide-semiconductor | **DHBT** – double heterojunction bipolar transistor | **GaAs** – gallium arsenide | **GaN** – gallium nitride | **HBT** – heterojunction bipolar transistor | **HEMT** – high electron mobility transistor | **InP** – indium phosphide | **ITU-R** – International Telecommunication Union Radiocommunication Sector | **LOS** – line-of-sight | **mHEMT** – metamorphic high electron mobility transistor | **MIMO** – multiple-input, multiple-output | **MMIC** – monolithic microwave integrated circuit | **MOSFET** – metal-oxide-semiconductor field-effect transistor | **NFmin** – minimum noise figure | **pHEMT** – pseudomorphic high electron mobility transistor | **QAM** – quadrature amplitude modulation | **SOI** – silicon on insulator | **SiGe** – silicon-germanium

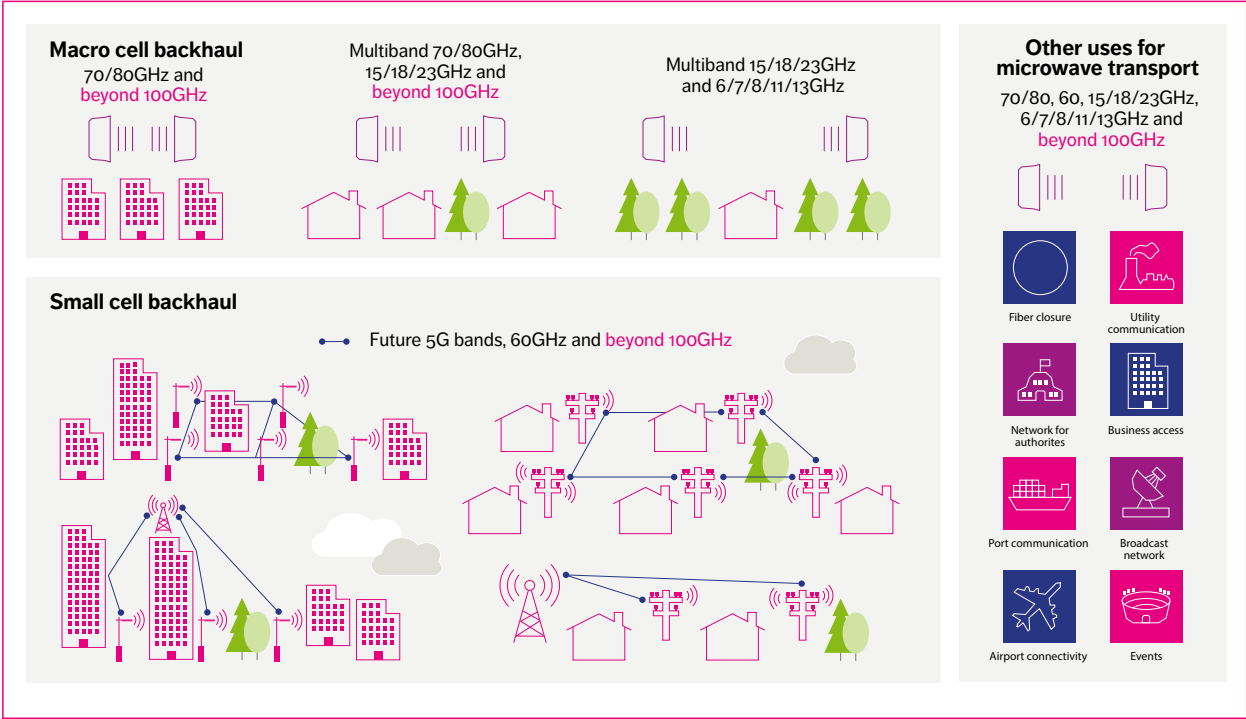


Figure 1
Future use of spectrum for microwave backhaul, including solutions beyond 100GHz

The spectrum above 100GHz consists of a multitude of sub-bands of different sizes with passive service allocations in between, as shown in **Figure 2**. The reason even wider continuous spectrum is not made available is to prevent interference with passive radiocommunication services such as the Earth Exploration-Satellite Service and the Radio Astronomy Service.

There is some interest in the use of frequencies beyond the D band for fixed service systems in the even longer term. Several frequency bands in the 275-1000GHz range have been identified for passive services, but this does not preclude their use for active services [1]. ITU-R will carry out studies until the World Radiocommunication Conference 2019 on the identification of frequency bands in the 275-450GHz range for land mobile radio and fixed

services applications [1]. It should be noted that the 252-275GHz frequency range is already allocated to fixed services. If 275-320GHz was added to this, it would form a continuous 68GHz wide band with moderate atmospheric absorption, as shown in **Figure 2**. This could be useful for fixed service applications in the distant future.

Attenuation due to atmospheric gases and rain [8] increases with frequency and there are also several absorption peaks, as illustrated in **Figure 2**. However, between the peaks, the attenuation increases quite slowly beyond 70GHz. For example, it increases about 2dB/km from 70GHz to the D band and about 4dB/km from 70GHz to 275GHz. The free space path loss [8] also increases with frequency: by about 6dB from 70GHz to the D band and about 11dB from 70GHz to 275GHz, for

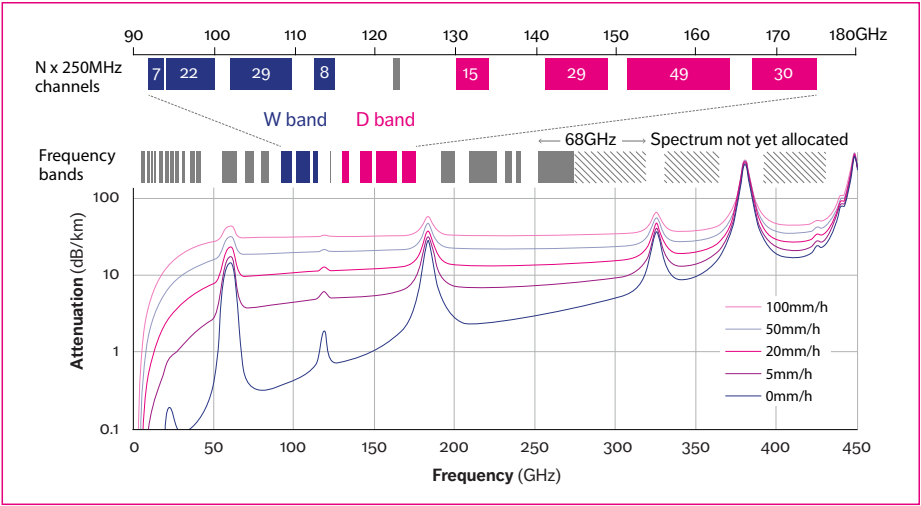


Figure 2
Frequency bands and atmospheric attenuation beyond 100 GHz

example. The propagation conditions are thus only slightly worse beyond 100GHz.

It is important for spectrum regulations beyond 100GHz to enable emerging and future innovations that can support capacities on the road toward 100Gbps. They should cover traditional link configurations, such as FDD, as well as complementary future innovations that might better handle the asymmetric and partly unpaired sub-bands, as illustrated in **Figure 3**.

Like fiber transport networks, microwave backhaul has historically been designed to be symmetrical. In most cases, the frequency bands are divided symmetrically into high and low sub-bands, used with FDD. Used to boost capacity and

IT IS IMPORTANT FOR SPECTRUM REGULATIONS BEYOND 100GHZ TO ENABLE EMERGING AND FUTURE INNOVATIONS

spectral efficiency, line-of-sight (LOS) multiple-input, multiple-output (MIMO) is an innovation that initially gained interest [4, 9], but has waned lately on account the more attractive multiband solutions. However, the small spatial antenna separation required for LOS MIMO in the D band makes it interesting on the road toward 100Gbps capacity. Multiband solutions, which enable enhanced data rates by combining resources in multiple frequency bands, constitute an essential part of modern radio access. As such, they have recently also become a topic of great interest in microwave backhaul [3] by making it feasible to use higher frequencies such as 70/80GHz over much longer distances. Multiband is also a very attractive option beyond 100GHz.

Today, the limited spectrum with unpaired or asymmetric sub-bands is used with TDD. FDD with asymmetric channels has been studied, but deemed too complex and of limited value in existing symmetric bands [10]. Asymmetric multiband solutions might be of interest in unpaired spectrum, rather like supplemental downlink for radio access. Flexible FDD configurations use separate transmit and receive

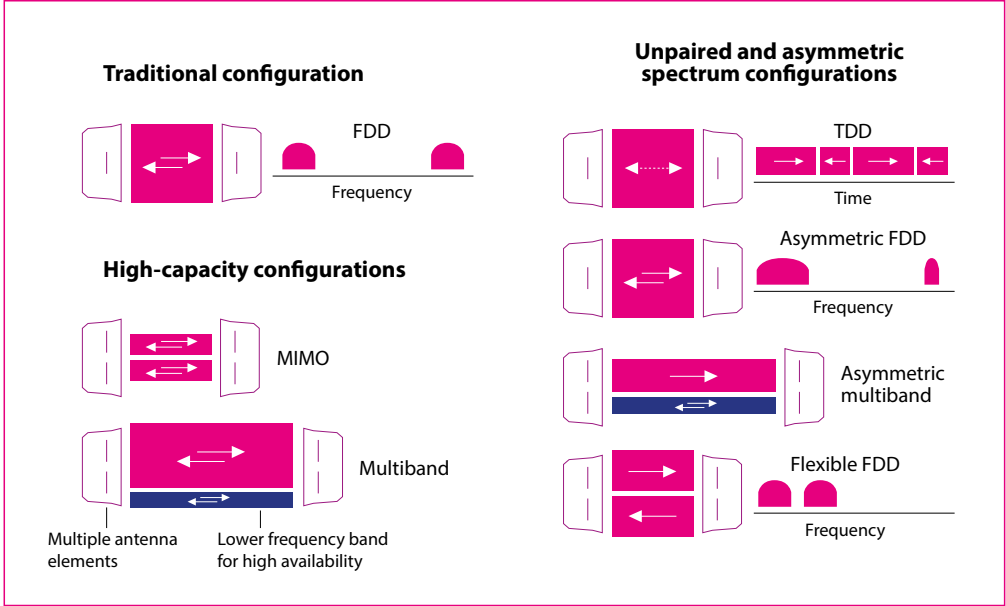


Figure 3
Examples of potential configurations beyond 100GHz, to support high capacities and facilitate use of unpaired and asymmetric spectrum

antennas instead of diplex filters for isolation [5, 6]. This does not add any spectrum efficiency, but might provide for better performance than that enabled by TDD in unpaired spectrum.

The road to 100Gbps transport solutions

Microwave backhaul technology has evolved tremendously in recent decades, repeatedly exceeding capacity limits and reaching performance levels only believed possible for fiber solutions. The commercial 70/80GHz equipment that is currently being introduced supports 10Gbps in 2GHz channels (8 x 250MHz) and it is reasonable to expect 20Gbps solutions in the future. Higher capacities are facilitated by wider channels, but national spectrum administrations commonly limit the maximum allowed channel size to secure a fair division among different users. The maximum channel size is typically limited to about 10 percent of the total band. For higher frequency

spectrum, with a greater possibility of frequency reuse, channels of up to about 20 percent of the total band may be allowed.

Realistic solutions on the continued road towards 100Gbps in different frequency bands are shown in **Figure 4**. Even wider channels up to about 5GHz (20 x 250MHz) might be obtainable in the D band, enabling solutions supporting 20Gbps, 40Gbps and even up to 100Gbps in the longer term, as indicated by the diamonds in Figure 4. But there are many technology challenges on this road, such as transmitter noise, signal distortion and other impairments that might limit maximum modulation order for extremely wide channels. Higher capacities and wider channel bandwidths also place more requirements on digital data converters. More advanced solutions using dual polarization – and even LOS MIMO – would enhance capacity but they also add cost.

The use of LOS MIMO solutions beyond

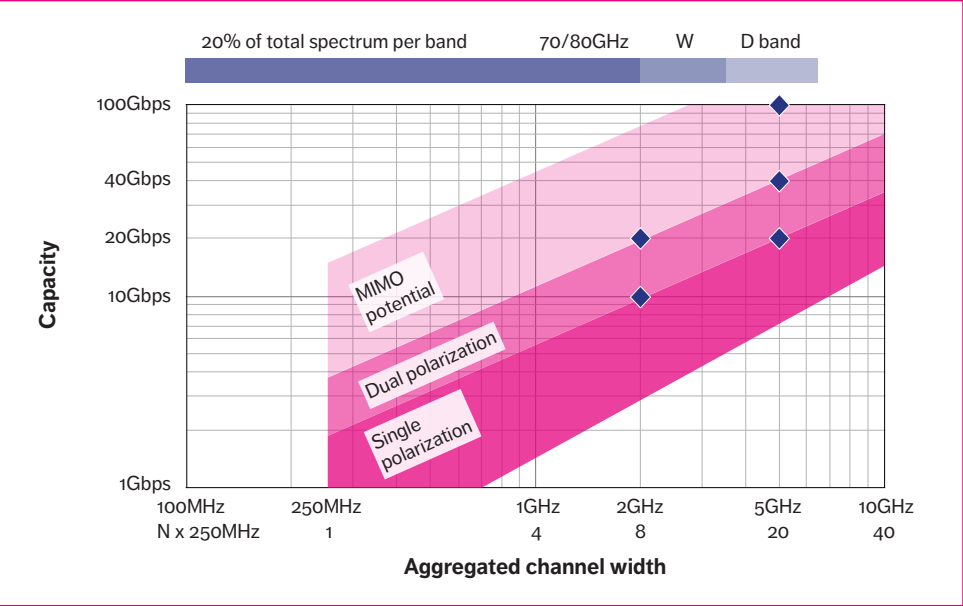


Figure 4
Realistic capacity versus channel bandwidth with single polarization, dual polarization and MIMO

100GHz carrier frequencies is attractive due to the reduction in required spacing between the antenna elements as the frequency increases. The optimal antenna separation d_{opt} , in a vertical and horizontal direction, may be written as [11]:

$$d_{opt} = \sqrt{\frac{cD}{fN}}$$

Where f is the frequency, c is the speed of light, N is the number of antenna elements in the vertical or horizontal direction and D is the hop length. A separation of 70-80 percent of the optimal value is possible, with only a limited decrease in system gain [9]. For example, at 155GHz, an antenna separation of 0.4m would be needed for a 300m hop distance, and 0.8m for a 1km hop. There are technological challenges (such as signal processing) involved in developing LOS MIMO in the D band, but in the longer term it is expected to enable the

final step to 100Gbps capacities, and even beyond, as illustrated in Figure 4.

Hop lengths beyond 100GHz

When assessing the ability of microwave backhaul to provide high-capacity transport over distance, three parameters should be considered:

- » the total system gain – the transmitted power plus the antenna gains minus the required received signal power
- » the targeted availability – the accumulated time a selected capacity should be sustained over the hop, which is usually expressed in a percentage of time per year, where 99.99-99.999 percent are common telecom grade targets
- » the local climate – the hop planning is done with propagation prediction methods using long-term rain and multipath statistical data for the hop location

The maximum hop length versus total system gain for differing levels of availability and local climate

SEMICONDUCTOR TECHNOLOGIES FOR BEYOND 100GHZ USE HAVE UNDERGONE A TREMENDOUS EVOLUTION IN THE PAST FEW DECADES

conditions at 155GHz is shown in Figure 5. It illustrates the total system gain for two equipment examples: one with 50dBi antennas, which is the general recommended maximum antenna gain in practical microwave deployments; and one with 35dBi, which is the recommended maximum antenna gain for sites with mast sway, such as small cell backhaul sites mounted on lighting poles. Each of the examples is for configurations supporting the 10 to 100Gbps examples in Figure 4, which all have similar system gains. As D band technology is maturing, transmitted power and receiver sensitivity of the same order as for today’s 70/80GHz equipment are expected, even if early implementations might have much lower system gain, as illustrated in Figure 5.

The 20, 50 and 100mm/h rain rates, exceeded for 0.01 percent of time per year, are representative for mild, moderate and severe local climate conditions. The availabilities of 99.9 percent and 99.995 percent in Figure 5 correspond to a propagation loss that exceeds the total system gain for about 9 hours/year and of 26 minutes/year. Using adaptive modulation, a lower modulation level in heavy rain increases the system gain to avoid transmission errors, but results in reduced capacity. For example, reducing modulation from 64QAM to BPSK correspond to 15dB increase of system gain, but a reduction to 17 percent of capacity. As Figure 5 illustrates, hop lengths of a few hundred meters are achievable for lower gain antennas. Using high gain antennas, it is possible to achieve hop lengths of about 1-2km and even up to 2-4km

for lower availability targets, such as multiband configurations. The hop lengths in the D band are thus well suited for urban and suburban deployments.

Semiconductor technologies as key enablers
Semiconductor devices are essential in all modern radio technology. Microwave backhaul equipment has historically relied on gallium arsenide (GaAs) circuits. More recently, gallium nitride (GaN) has been introduced in commercial products due to its high breakdown voltage enabling higher transmit power. There is also considerable interest in silicon chipsets, based on CMOS or SiGe-HBT, due to their lower production cost per chip in high volumes and high integration density. These are particularly relevant for short range deployments where high output power is less important, such as in the 60GHz frequency band.

Driven by the space, defense and imaging industries, semiconductor technologies for beyond 100GHz use have undergone a tremendous evolution in the past few decades [12]. There are today a few commercial technologies available for beyond 100GHz applications and several more are being researched for even higher performance, as shown in Figure 6. The three main transistor technology classes are HBT, HEMT, and MOSFET [12], where MOSFET is typically implemented in SOI CMOS for high frequency operation. A key property is the feature size, since a transistor with smaller feature size supports higher frequencies. As a rule of thumb circuits are designed to operate at below half fMAX, where fMAX is the frequency at which the transistor’s power gain is equal to one. It is possible to bring the operation frequency much closer to fMAX but doing so results in lower energy efficiency and higher design costs. Other important material properties are the minimum noise figure (NFmin) and the breakdown voltage (Vbr), which determine receiver sensitivity and maximum transmitted power, respectively. The right column in Figure 6 indicates the commercial maturity of the technology, where additional aspects are the development and production cost. Flicker noise

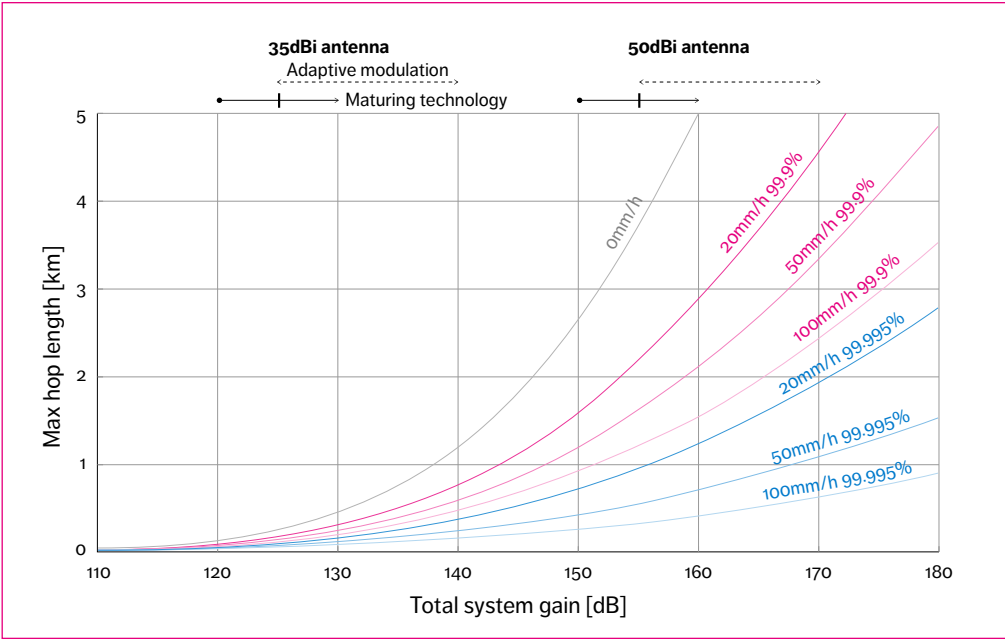


Figure 5
Maximum hop length versus total system gain at 155GHz, for different rain intensities (exceeded 0.01 percent of the year) and for two different antenna configurations

Technology	Feature size (nm)	fMAX (GHz)	Vbr (V)	NFmin (dB) at 50GHz**	Production or research?
GaAs pHEMT	100	185	7	0.5	P
GaAs mHEMT	70	450	3	0.5	R*
GaAs mHEMT	35	900	2	1	R
InP HEMT	130	380	1	<1	R
InP HEMT	30	1200	1	<1	R
GaN HEMT	60	250	20	1	R
GaN HEMT	40	400	42	1.2	R
SOI CMOS	45	280	1	2-3	P
SiGe-HBT	130	400	1.4	2	P
InP DHBT	250	650	4	3	R*
InP DHBT	130	1100	3		R

*Ready to be commercialized in 1-2 years
**NFmin is proportional to the frequency.

Figure 6
Overview of semiconductor technologies beyond 100GHz and their key parameters

RESEARCH ON HIGH-FREQUENCY TECHNOLOGIES IS GAINING GLOBAL INTEREST

generation, memory effects and temperature behavior are not included in the table, but should also be considered.

The maximum transmitted power limits the system gain. Research has been published on power amplifiers in GaAs, InP and SiGe technologies delivering more than 10dBm of output power beyond 100GHz [13-15]. GaN is in the future expected to demonstrate even higher output power due to the materials high breakdown voltage. GaAs pHEMT provides high breakdown voltage and a low noise figure and, in a few years, is also expected to be able to support the D band. InP supports very high frequencies, albeit at a high material cost. Because of its good performance it could be useful for research and predevelopment activities of equipment in the D band. It might also be applicable for longer term commercial applications around 275GHz.

Silicon technologies such as SOI CMOS and SiGe-HBT are today feasible up to the D band although the maximum output power is limited due to the low breakdown voltage of silicon and the noise figure is worse compared to GaN and GaAs technologies. Due to the excellent properties for high integration, silicon technologies are promising for short-range, low-cost applications beyond 100GHz.

There are many additional obstacles to overcome. Packaging and interconnect above 100GHz are challenging due to the short wavelengths. Parasitic effects are more pronounced and the tolerance requirement is high in design, manufacturing and assembly, especially when considering wide bandwidths. Crosstalk and unwanted resonances are additional issues since the typical monolithic

microwave integrated circuit (MMIC) size is of the order of the wavelength. This makes traditional interconnects, such as wire bonding and flip chip, difficult to use with high yield.

Research on high-frequency technologies is gaining global interest. One example is the non-galvanic chip-waveguide interconnects currently being investigated by the European Union funded Horizon 2020 project M3TERA, where low-loss silicon waveguides are made using a 3D micromachining technique that provides a silicon platform with embedded components for industrialized assembly. Another example is the research program commissioned by Japan's Ministry of Internal Affairs and Communications, "R&D Program on Multi-terabit Gigabit Wireless Communication Technology at Subterahertz Frequencies," which investigates radio sources beyond 275GHz. A third example is the Horizon 2020 funded research program TWEETHER, which focuses on high-power amplifiers beyond 100GHz.

It is a long and winding road from research to full fledge commercial equipment that meets the right performance and cost. Ultimately this can only be achieved with a competitive industry eco-system sharing a common vision [6].

Putting theory to the test

Working with researchers at Chalmers University of Technology in Gothenburg, Sweden, Ericsson Research has developed a D band transceiver module, shown in **Figure 7**. The module contains an InP DHBT MMIC and a separate circuit board for bias control and connectors. The MMIC covers the entire D-band. The red square in the photo shows the location of the MMIC, which measures 1.3mm x 0.9mm. The close-up on the right shows the transceiver MMIC glued to a silicon carrier and connected to the module with wire bonds.

Both transmitter and receiver MMICs contain a Gilbert cell mixer for up or down conversion and a frequency tripler for local oscillator generation. A low-noise amplifier is implemented in the receiver

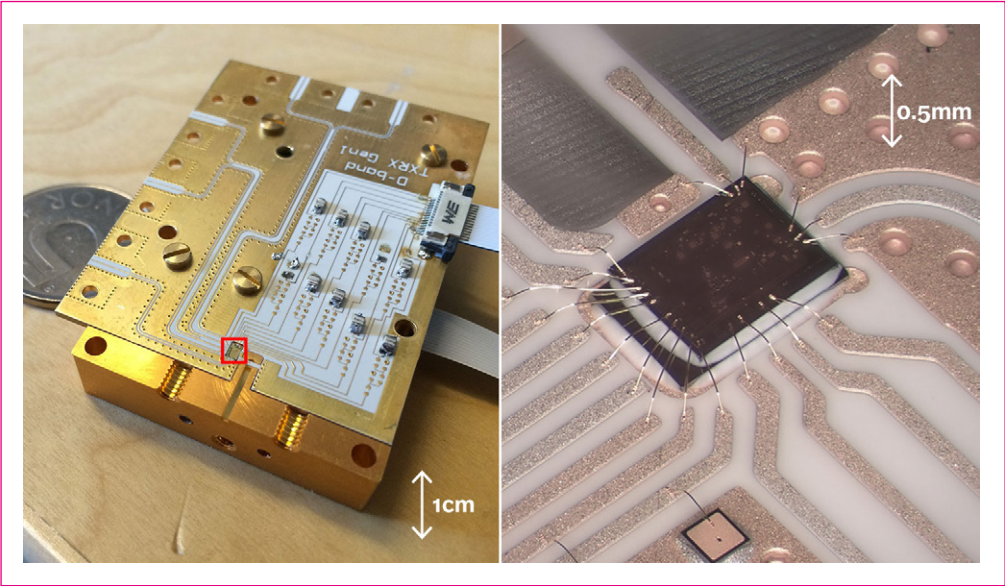


Figure 7
D band transceiver module (left) with a red square indicating the position of the wire-bonded InP DHBT transceiver MMIC (shown in close-up on the right)

MMIC having approximately 15dB of gain, while a medium-power amplifier is implemented in the transmitter MMIC supporting a saturated output power of more than 10dBm [15]. The MMICs are assembled in a slot inside a 50µm thick soft substrate that also extends into a waveguide as an E-plane probe. The waveguide connects to a diplex filter that interfaces with an antenna.

The transmitter and receiver modules were measured back-to-back before being assembled into the radio prototype. **Figure 8** shows the measured bit error rate (BER) versus received signal power for a 125MHz channel at 143GHz. The modules supported up to 5GHz channels and the inset in **Figure 8** shows the measured error-free constellation for a symbol rate of 4GBaud using 16QAM for in total 16Gbps [15]. A noise figure of 9.5dB was measured for the receiver MMIC, which is a good result for receiver chipsets based

on bipolar technologies at these frequencies. The 10⁻⁶ BER threshold of -63dBm for 4QAM (in **Figure 8**) indicates that these early transmitter and receiver modules add a penalty of more than 8dB to the receiver sensitivity. These results emphasize the need for careful control of how the module is designed and built.

The photo on the left in **Figure 9** shows the complete radio prototype mounted in an enclosure together with the modem and antenna for outdoor over-the-air measurements. The antenna is only 7.5cm in diameter, but still provides 40dBi gain. Long-term tests on frequencies beyond 100GHz will be important to validate the ITU-R propagation and availability models, similar to what was initially done in the 70/80 GHz band [16]. The small antenna footprints at these high frequencies could enable new compact radio concepts, as illustrated to the right in **Figure 9**.

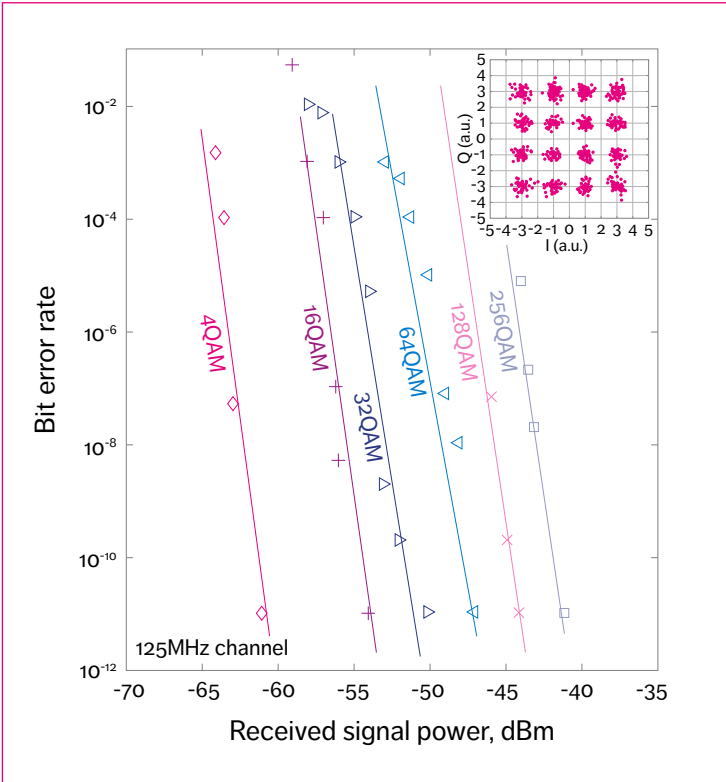


Figure 8
Measured bit error rate at 143GHz versus received signal power. Inset shows measured constellation diagram at 4GBaud and 16QAM modulation for in total 16Gbps



Figure 9
D band radio prototype (left) and visionary design idea (right)

Conclusion

The ceaseless quest to provide higher data-carrying capacities has led to the use of ever higher frequencies where more spectrum is generally available. The tremendous growth in the use of the 70/80GHz band that we can see today was made possible by several years of research and development and a great deal of work on spectrum regulation, as well as the experience gained from several technology and product generations. Similar efforts are now underway on the road to microwave backhaul beyond 100GHz, supported by the rapid evolution of high frequency semiconductor technologies and promising new devices. In light of this, we expect to see the large-scale deployment of beyond 100GHz solutions in 2025 to 2030. The W and D bands will undoubtedly be able to support capacities in the 5 to 100Gbps range, over distances up to a few kilometers. *

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Securing the cloud

WITH COMPLIANCE AUDITING

More and more companies are moving their applications and data to the cloud, and many have started offering cloud services to their customers as well. But how can they ensure that their cloud solutions are secure?

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Security compliance auditing is an assessment of the extent to which a subject (a cloud services provider or CSP, in this case) conforms to security-related requirements. At a minimum, a CSP must be able to deploy tenants' applications, store their data securely and ensure compliance with multiple regulations and standards.

■ Many industry sectors – healthcare and utilities, for example – are highly regulated and have to meet stringent data privacy and protection requirements. To serve these types of companies, cloud providers must be able to prove their alignment with the latest standards and regulations such as the Health Insurance Portability and Accountability Act (HIPAA), the Payment Card Industry Data Security Standard (PCI DSS) and the Federal Risk and Authorization Management

Program (FedRAMP). Without the right set of tools in place, cloud characteristics such as elasticity, dynamicity and multi-tenancy make proving compliance with such standards both challenging and costly.

Regulations such as HIPAA and PCI DSS define auditing and proving compliance with industry standards and regulations as shared responsibilities. To address users' compliance-related needs, cloud providers must demonstrate evidence of compliance with regulatory requirements across industry segments.

Figure 1 illustrates the cloud security compliance landscape. Providers that can offer tenants credible, trustworthy compliance information on relevant requirements at any time, in a cost-efficient manner, stand to gain a significant competitive advantage.

Auditing security compliance typically involves the manual inspection of regularly generated

audit reports and logs, and possibly dynamic tests conducted at runtime. However, applying such techniques in the cloud would be time consuming and costly owing to cloud characteristics.

For instance, to prove network isolation, all layers such as cloud management as well as the virtual network, overlay network, real network (non-virtual), and physical network have to be verified. The results of each verification process on the layers are correlated to avoid any gaps. Current practices such as design document verification, network traffic injection and penetration testing don't work in an environment where tenants share resources, and network parameters change quickly and dynamically.

Operators and cloud providers therefore need a new set of automated tools and techniques that can manage security and compliance, protect consumers' assets, and enable security-related services – in a continuous and cost-effective fashion. In telecom context, the European Telecommunications Standards Institute (ETSI) has proposed an architecture for continuous security monitoring and lifecycle management for network function virtualization to satisfy security requirements at both the operator and consumer level [1].

The ways in which evidence of compliance is provided in the cloud marketplace vary widely at

CLOUD PROVIDERS MUST DEMONSTRATE EVIDENCE OF COMPLIANCE WITH REGULATORY REQUIREMENTS ACROSS INDUSTRY SEGMENTS

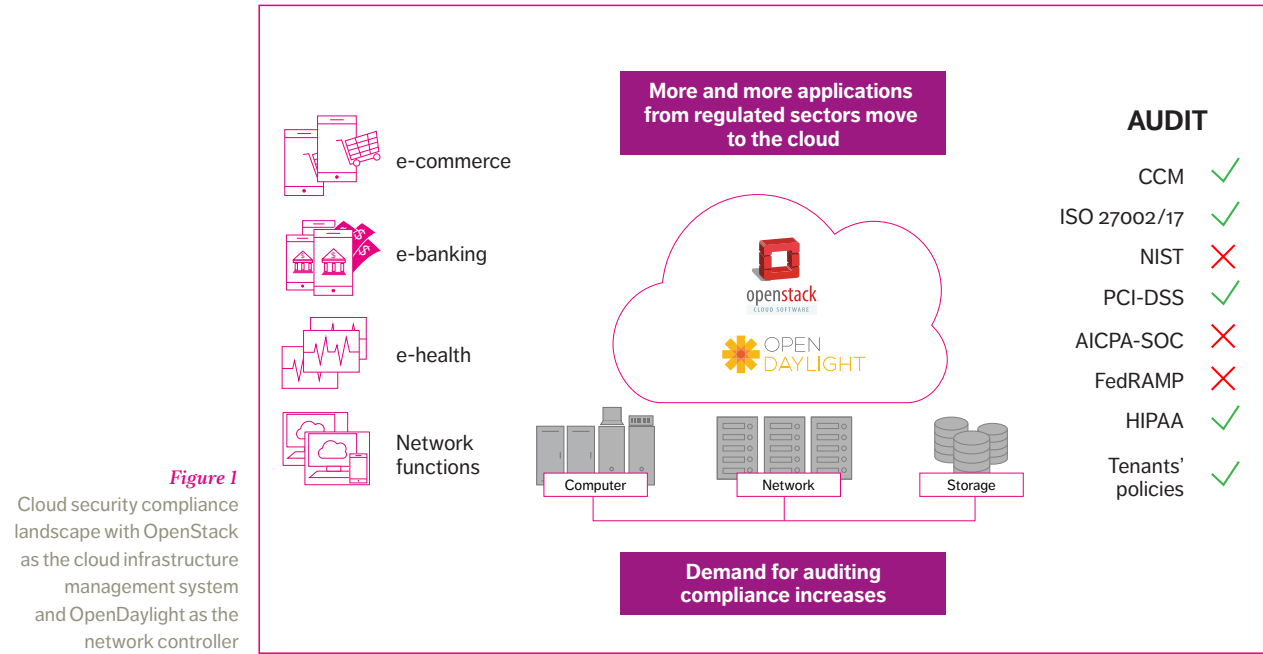
present. It is problematic for a tenant to evaluate cloud providers' capabilities and to understand which party is responsible for what from a compliance perspective. Trust between tenants and their providers is often based on legal texts and disclaimers that can be difficult to comprehend. There is clearly room for improvement, as evidenced by the European Union's call for closer adherence to privacy regulations by global CSPs.

Compliance standards in the cloud

To ensure compliance with different security frameworks in the cloud, there are two main types of standards: vertical and horizontal. Horizontal standards are generic standards that are applicable to many industries. Vertical standards are applicable to specific industries. Several standards (horizontal and vertical) have been supplemented

Terms and abbreviations

AICPA – American Institute of Certified Public Accountants | AWS – Amazon Web Services | CCM – Cloud Controls Matrix | CCS – Control Compliance Suite/Services | CSA – Cloud Security Alliance | ETSI – European Telecommunications Standards Institute | FedRAMP – Federal Risk and Authorization Management Program | GRC – governance, risk management and compliance | HIPAA – Health Insurance Portability and Accountability Act of 1996 | IaaS – infrastructure as a service | ISO 27001 – specification for an Information Security Management System (ISMS) | ISO 27018 – code of practice for protection of personal data | NIST – Network Information Security & Technology | NIST SP – Network Information Security & Technology Special Publication | NoSQL – not only Structured Query Language | PaaS – platform as a service | PCI DSS – Payment Card Industry Data Security Standard | SaaS – software as a service | SIEM – security information and event management | SOC 1, 2, 3 – Service Organization Controls type 1, 2, 3 report | SQL – Structured Query Language | V&V – verification and validation | VM – virtual machine



to guide certification handling in the cloud computing domain.

Besides the establishment of horizontal and vertical standards by standardization bodies, other organizations and informal groups such as the Cloud Security Alliance (CSA) address standardization issues related to cloud computing and work on promoting best practices and reaching a consensus on ways to provide security assurance in the cloud. For example, the CSA's cloud security governance, risk management and compliance (GRC) stack [2] supports cloud tenants and cloud providers to increase their mutual trust and demonstrate compliance capabilities.

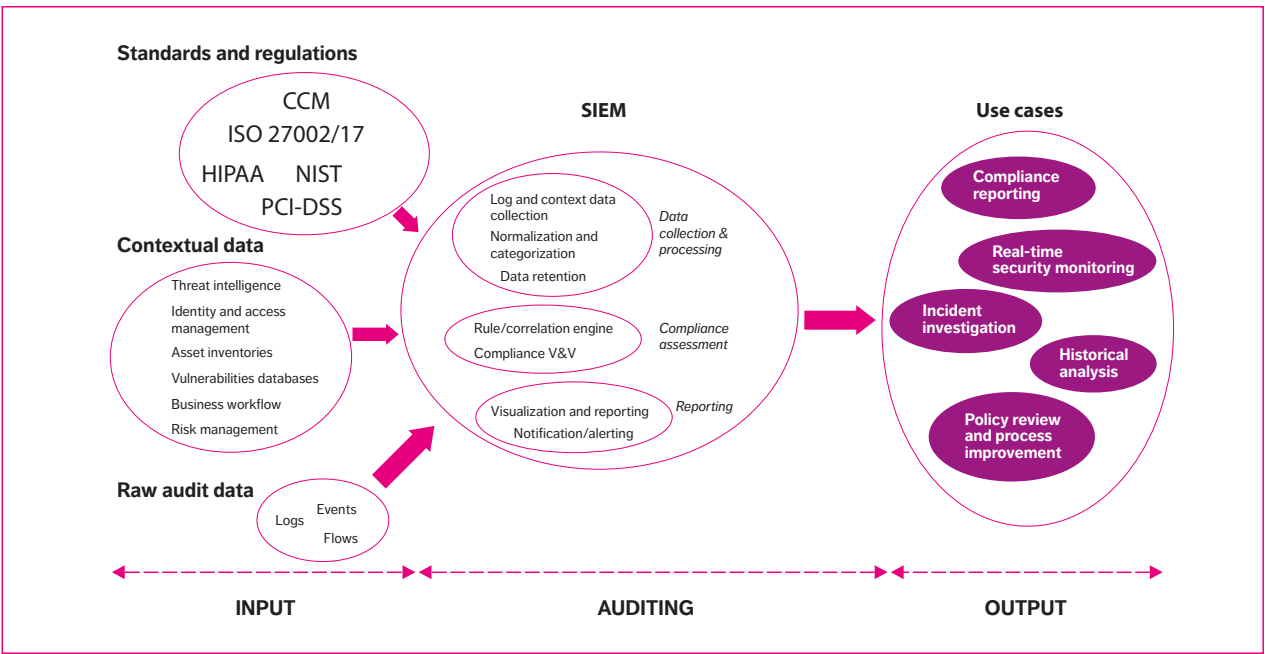
Current auditing tools

The auditee – in this case the cloud provider or consumer – is required to produce compliance reports to prove that their security measures

are protecting their assets from being compromised. Additionally, regulatory bodies require the auditee to retain log data for long periods of time, making it possible for auditors to analyze audit trails and logs. To this end, the auditee can use different types of tools to manage and maintain a holistic view of the security of its environment.

Several open source and commercial tools, including security information and event management (SIEM) and GRC tools, that enable generation of compliance reports on a periodic and/or on-demand basis, exist in the market. **Figure 2** illustrates the main input, output and functionality of an SIEM tool.

In addition to SIEM functionality, GRC [3] tools deliver the core assessment technologies to enable security and compliance programs and support IT operations in the data center.



They enable information security managers to address IT governance, risk and compliance issues by helping them to prevent and respond to non-compliance of security controls while taking into account tolerated risk.

Enterprise class tools

With the advent of the cloud, the makers of several enterprise class tools have proposed integration of their solutions into the cloud environment. While many enterprise-class SIEM engines rely exclusively on correlation to analyze audit data, a new generation of cloud-specific tools includes log search engines and advanced analytics to process the large amount of data and gain security intelligence and knowledge. Nonetheless, most of these tools have been designed to work in enterprise environments whose characteristics differ significantly from the cloud.

Open source projects

Due to the increasing importance of auditing and monitoring in the cloud, open source projects have been created as part of existing cloud management software. For example, OpenStack Congress aims to offer governance and compliance assurance by providing policy as a service. It targets IaaS and does not cover any PaaS or SaaS deployment. Specifically, it allows declare, audit and enforce policies in heterogeneous cloud environments.

A drawback of OpenStack Congress is that it does not allow a full verification through all layers – verification is limited to the information provided by OpenStack services. An elastic stack based on open source tools is another option. This alternative consists of a data search stack that encompasses several components, namely: Kibana for data visualization; Elasticsearch for searching, analyzing and storing data; as well as Beats and

Logstash for data collection from various sources in different formats.

When comparing commercial tools with these open source projects, a notable benefit of commercial tools is that they have most of the audit process ready to use out-of-the-box.

Cloud-based services offered by cloud providers

Some cloud IaaS providers are currently proposing partial solutions to help consumers verify that their applications are handled in conformance with their security policies. For instance, AWS offers dynamic customizable compliance checking of cloud resources using AWS configuration rules. Other tools have also been proposed, such as Inspector by Amazon, which is an automated security assessment service that finds security or compliance issues on applications launched within AWS instances.

Cloud-specific tools

Cloud-specific tools such as Catbird Secure offer policy compliance automation and monitoring solutions for private and hybrid cloud environments and focus on software-defined security. Another example is RiskVision Continuous Compliance Service (CCS), an on-demand service allowing providers to gain visibility into their cloud risk exposure and to manage compliance.

Challenges and implementation gaps

A number of challenges make techniques for auditing conventional IT systems unsuitable for use in a cloud environment without significant adaptation. While several common concerns arise when auditing in both domains, a cloud security audit must address unique problems.

Compliance responsibility

Cloud applications run in different deployment models (IaaS, PaaS, or SaaS) and on different types of cloud (public, private or hybrid). This rich set of combinations leads to a complex control dependency and complicates the responsibilities of different actors. The reliance on the CSP varies

according to the deployment and type of cloud. For example, in a public IaaS, the hardware and virtual layers are managed by the CSP while the application layer is managed by the tenant. Therefore, there is limited reliance on CSP in IaaS, but most reliance on CSP in SaaS. Thus, it is necessary to define a clear model for the shared responsibility of compliance management.

The massive size of the cloud

The large scale of cloud environments – with the increased number of virtual resources and sources of data – has a direct impact on the size of audit trails and logs. Given the huge amount of data held in them, efficient collection, manipulation and storage techniques are required. Conventional tools were not conceived for large-scale data – they use off-the-shelf fixed-schema SQL databases with a centralized system for the analysis of audit trails. The scale and performance limitations of this type of architecture represent a single point of failure. Auditing and compliance verification tools for the cloud must be designed from scratch to process a very large quantity of data while meeting performance requirements.

The rapidity and dynamicity of cloud services

The speed of events and operations in the cloud constantly changes logs and configuration data. For example, each time a new virtual machine (VM) is created or migrated, new data is generated that may change the compliance status. This is becoming more complex as cloud providers are moving toward more real-time programmable controls by using software-defined networks and NFV in their cloud data centers. One of the major issues in conventional solutions is that they are conceived to execute in a quasi-static environment where auditing is generally performed periodically and remains valid until the next period. They mainly verify a snapshot of the security state at the time of the audit. This is not sufficient in the cloud, where audit and compliance assurance is required each time the infrastructure changes to assess whether these changes give rise to security gaps or infrastructure misuse.

If an audit and compliance assessment tool cannot cope with the high rate of configuration changes for large data centers, it is not fit for the task. Changes in the cloud require the ability to automatically collect data to present near-real-time visibility about compliance to tenants and auditors alike.

Multi-tenancy in the cloud

Audit trails and logs are currently being generated for different actors (tenants, users, cloud provider and so on) on shared physical and virtual layers without a clear separation between them. This approach cannot address all the needs rising from complex use cases such as when a cloud broker leases virtual resources to a third party. Furthermore, it may not be possible for auditing tools to monitor the full stack from the hardware layer up to the application layer because of potential compromise of the privacy of other tenants and of the confidentiality of sensitive information concerning the cloud infrastructure. This is why some providers (particularly SaaS ones) restrict vulnerability assessments and penetration testing, while others limit availability of audit logs and activity monitoring. Most conventional tools are simply not designed to support multi-tenant environments. Therefore, different accessibility schemas must be put in place to give the right access to the common logs for different tenants based on the roles and privileges of different actors.

Privacy protection and GRC support

A CSP with a multi-tenant environment is forbidden to reveal details or metadata that would compromise tenants’ privacy or security. Nor is it allowed to disclose any sensitive information to a third party and it must protect against attackers accessing any significant information about the tenants. At the same time, mandated auditors need to access useful and complete information to provide evidence of compliance. In addition, tenants need to receive the right assurances from the CSP and the auditors or perform their own compliance audit of their setting in the cloud,

●● **AUDITING AND COMPLIANCE VERIFICATION TOOLS FOR THE CLOUD MUST BE DESIGNED FROM SCRATCH TO PROCESS A VERY LARGE QUANTITY OF DATA WHILE MEETING PERFORMANCE REQUIREMENTS** ●●

independently of the cloud provider. Therefore, auditing tools should allow for securely outsourcing anonymized logs and audit trails to different interested entities without sacrificing privacy and sensitive information for an evidence-based audit and GRC approach in the cloud.

Trust and integrity of audit data

Audited data is often considered to be inherently reliable. But before being presented to the auditor, the original pieces of data will have been passed from the source to the presentation layer via communication interfaces and processed by dynamic software instances. The degree of trust in such a chain is hard to evaluate. Many cloud solutions enable an assessment of the trustworthiness of the hardware platform and bootstrapping of the virtual machines, and safeguard the integrity of log files at rest and in transit. However, audit data would not necessarily be approved as evidence in court if the data integrity had been compromised during any step of the process. The integrity of the audit data source, of the data collector and of the log server should be attestable, assuming that appropriate controls are in place for securing the audit data itself and that there is proof of mutual authentication between the processing elements with an accepted security strength.

●● A CONTINUOUS COMPLIANCE VERIFICATION MODEL PROVIDING TENANTS WITH COMPLETE COMPLIANCE VISIBILITY IS KEY TO REDUCING AND LIMITING EXPOSURE TO RISKS ●●

Achieving truly effective auditing in the cloud

In light of the challenges to creating an effective auditing approach in the cloud using the conventional techniques, it is useful to highlight some of the key characteristics of an effective cloud auditing solution.

Continuous monitoring and high automation for compliance

As the cloud is inherently elastic and dynamic, an effective auditing framework must be augmented by continuous compliance and monitoring features [1]. This is not only necessary to maintain compliance but also to improve overall security. It must also provide a high level of automation to cope with quick and transparent changes in collaboration with the cloud management system. Automation is necessary to collect the right information in near real-time and from the right source. Additionally, to enforce an evidence-based compliance verification in a multi-tenant environment, the CSPs should expose information gathered from trusted monitored sources in an open standard format while protecting tenants' privacy by using, for example, anonymization of traces and audit trails for the auditors' and tenants' benefit. Therefore, moving towards a continuous automated compliance verification model that provides complete compliance visibility to the tenants is key to reducing and limiting exposure to risks related to compliance and security breaches.

Building auditing capabilities into the cloud infrastructure

It is much more effective and cost-efficient to build intrinsic auditing capabilities into the cloud infrastructure than to attempt to retrofit existing auditing approaches to the cloud environment. To provide various actors with the necessary audit trails without violating user and tenant privacy, the cloud infrastructure could implement labeling mechanisms to trace the logs to their target tenants. Tackling logs and audit trails in the cloud as opposed to a classical centralized log server in an enterprise environment requires a distributed log collection and retrieval mechanism. Building accountability and traceability into the cloud infrastructure is the best way to provide an efficient and effective auditing solution.

Using analytics for compliance verification

While conventional audit systems specialize in detecting known threats, providing support for identifying unknown threats is a new trend in auditing that is highly relevant to the cloud. Owing to the great quantity of audit data and logs in large data centers, the use of big data analytics based on data mining, machine learning and behavioral monitoring techniques for cloud auditing tools and SIEMs is increasing. In the same vein, storing raw audit data requires new database architecture and technology (such as NoSQL) or support of flat file databases. For the sake of scalability, new deployment options are being considered to move from centralized audit analyses to distributed ones. Analytics must be further explored and improved to tackle cloud-specific characteristics and their actual potential must be investigated in real-world deployments.

Modular compliance approach

Many cloud applications are deployed for highly regulated industries with different compliance needs such as PCI/DSS, HIPAA, ISO 27017 and ISO 27001. These compliance frameworks correspond to different security requirements, which in turn necessitate a large set of controls that must be put in place in the cloud infrastructure.

There are, however, many commonalities between the requirements of all these frameworks in terms of data storage obfuscation, data storage integrity and access control, for example. Therefore, a baseline security requirement needs to be defined to cover the major common requirements. This baseline should be augmented dynamically in the cloud to provide support for different compliance frameworks. Consequently, an efficient auditing approach should be modular, supporting the common denominator requirements as a baseline security requirement and adding different control modules to support specific security frameworks. The CSA CCM compliance matrix is a good starting point for aggregating the major common security requirements.

Application to 5G

5G networks are expected to play a central role in providing a common backbone for information exchange between various applications that belong to different industry segments, which would

mean that the security of these applications would depend on the security of the 5G network [4]. This would result in the need to certify 5G networks against all (or at least parts of) the security standards that are related to the served verticals. Implementing isolated network slices for different types of applications would ease compliance assurance by confining certification efforts to each single slice against the appropriate subset of the security requirements. Figure 3 shows one way this could be accomplished.

Conclusion

The cloud has become a standard in modern computing, and companies in many industry verticals are moving their data to it. Therefore, security assurance, auditing and compliance in the cloud is gaining momentum. Unfortunately, several challenges related to the particular specificities of cloud are limiting the potential benefit of applying current auditing practices and tools.

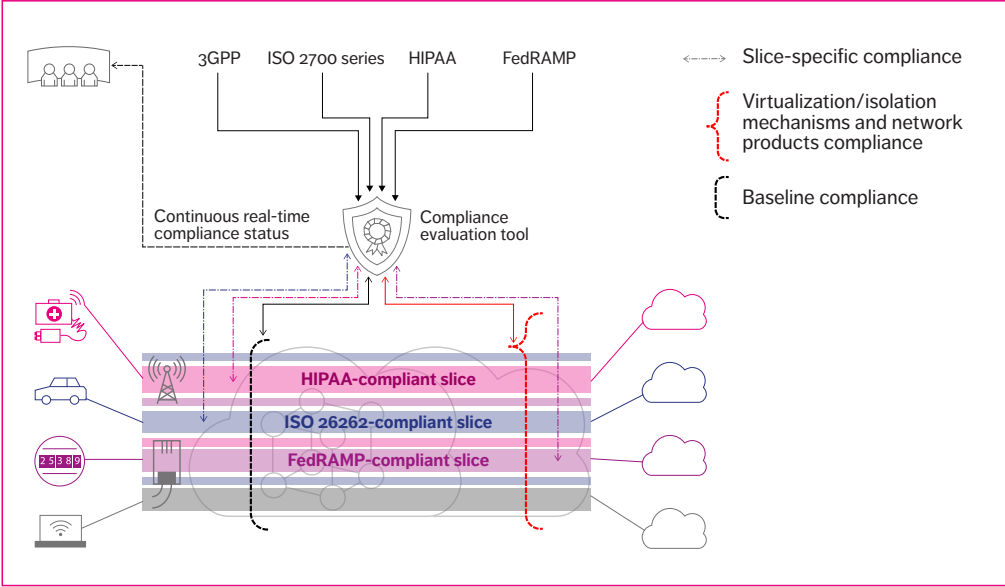


Figure 3 Application to 5G security compliance auditing

Moving toward a continuous automated compliance verification model that provides tenants with complete compliance visibility is key to reducing and limiting exposure to security-related risk. An effective and efficient cloud auditing solution must:

- » support large-scale cloud environments
- » offer a high level of automation
- » allow for near-real-time compliance visibility without compromising stakeholders' privacy and the confidentiality of sensitive data
- » fully support multi-tenancy
- » provide modular compliance verification to address several standards.

In light of these requirements, new auditing solutions adapted to the cloud environment must be proposed. *

Further reading

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- » S. Majumdar, Y. Jarraya, T. Madi, A. Alimohammadifar, M. Pourzandi, L. Wang and M. Debbabi, Proactive Verification of Security Compliance for Clouds through Pre-Computation: Application to OpenStack, 21st European Symposium on Research in Computer Security (ESORICS 2016), Heraklion, Greece, September 28-30, 2016, available at: https://link.springer.com/chapter/10.1007/978-3-319-45744-4_3

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TECHNOLOGY TRENDS DRIVING INNOVATION

– *five to watch*

Our industry has an increasingly important role to play in creating the foundation for new business in a broad range of industry sectors in countries all around the world. As Ericsson’s new Chief Technology Officer, it’s my job to keep track of technological advancements on the horizon and leverage them to create new value streams for society, consumers and industries. The challenge is timing, and to see new things in the context of the present without losing sight of history.

I have selected the five trends presented here based on my understanding of the ongoing transformation of the industry, including rapid digitalization, mobilization and continuous technology evolution, and how they affect the future development of network platforms – one of the essential components of the emergent digital economy. At Ericsson, our role is to keep these top trends in sight to guide our innovation, test our limits and ultimately create a thriving market for the next generation of technology. ➔

BY ERIK EKUDDEN, CTO

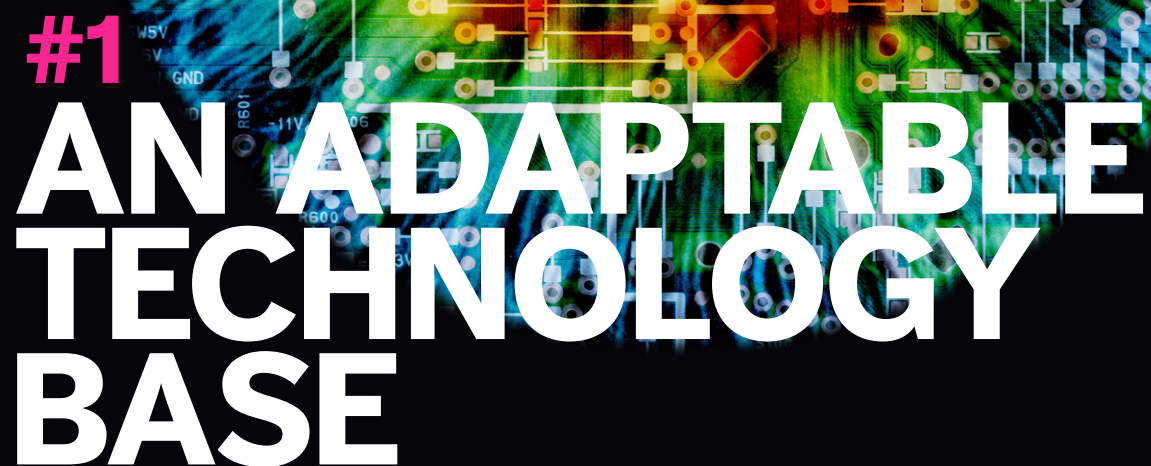
#1
**AN ADAPTABLE TECHNOLOGY
BASE**
Blending technologies in new
ways to unleash next generation
computational networks

#2
**THE DAWN OF TRUE MACHINE
INTELLIGENCE (MI)**
Moving from cognitive MI toward
augmented human intelligence

#3
**END-TO-END SECURITY AND
IDENTITY FOR THE INTERNET
OF THINGS (IOT)**
A holistic approach to trust in all
dimensions

#4
**AN EXTENDED DISTRIBUTED
IOT PLATFORM**
Acceleration toward a distributed
and connected IoT platform

#5
**OVERLAYING REALITY WITH
KNOWLEDGE**
Immersive communication
that ties user experience to the
physical world



#1 AN ADAPTABLE TECHNOLOGY BASE

TO RELEASE the full potential of the digital economy, the underlying technology components will rely on a symbiotic evolution in the software and hardware dimension. It is in the lowest layers of the technology stack – at the intersection between software and hardware – where more powerful and flexible solutions will become available, enabled by virtualization technologies and horizontal architectures.

ARCHITECTURAL ADJUSTMENTS AHEAD

Geometrical scaling has long been the main path for the transistor technology evolution but extensive research is currently underway to find alternative methods to increase transistor performance. Significant efforts are being put into building more advanced architectural structures, such as various types of non-monolithic integration technologies, to increase integration and thereby maintain the performance evolution track.

In the computing domain, the dominating CPU architecture trend is massive multicore to meet parallel processing demands. With more processors on a chip, memory architectures and data transfer will become key technologies in hardware. Non-volatile memory and integrated silicon photonics will reach maturity and change the entire memory/storage hierarchy. The new technologies are

expected to have significantly higher performance as well as lower latency and energy consumption.

To further enhance computational power and performance in data centers, specialized resources such as smart network interface controllers, general purpose graphics processors and field programmable gate arrays will be made available for virtual applications through abstractions. A similar co-processing architectural approach can be expected for quantum computing.

The rapid advances of base components will mean that the first exascale systems with computing power at double the capacity of today's top 500 computers combined can be expected within five years.

In combination with specific types of algorithms and applications, these technology shifts could prove to be disruptive.

SPURRING INNOVATION THROUGH ALGORITHM EVOLUTION

Massive data collection from, for example, IoT sensors will drive the need for new predictive software algorithms that also take advantage of the increasingly parallel computational power. Algorithm development will play an even more significant role in software design. One example is deep learning, a branch of machine learning that uses a layered algorithm structure to learn hierarchical concepts.

The amount of code needed to reach a higher level of complexity is very small, a reduction by a factor of 10 or more.

compared with traditional software system approaches. This type of system learns from examples: it utilizes a generic algorithm that uses the examples to set parameters in the algorithm to fit the particular task at hand.

Reinforcement learning is a technology to develop self-learning software agents, which can learn and optimize on an observed state of the environment and a reward system. This enables development of self-learning systems that require neither human intervention nor hand-engineered, threshold-based policies.

ENABLING THE FUTURE COMMUNICATION SYSTEM

From a connectivity perspective, one interesting area is beamforming in future 5G networks, where symbiosis of software and hardware plays an important role. At mm-wave frequencies, hundreds of antennas and transceiver chains work together with advanced control algorithms to generate, form and steer radio waves in real time to accomplish multiuser MIMO (multiple-input, multiple-output). Each device is accessed by a single user dedicated beam to optimize network and user capacity, efficiency and quality.

At Ericsson, we are currently investing in a variety of collaborative efforts with academia and the technology industry to foster an open environment in which to share ideas and visions that will enable the most successful future network from both a societal and individual human perspective. 🌐

#2 THE DAWN OF TRUE MACHINE INTELLIGENCE

ARTIFICIAL INTELLIGENCE (AI) first emerged in the early 1950s when symbolic logic and rule-based systems were used to generate logical conclusions. Another important step included artificial neural networks, capable of performing pattern recognition by learning from data through iterative optimization. The advancements in these computational-intense algorithms have progressed in performance and cost through the evolution of computing power, data availability and connectivity.

Machine intelligence (MI) combines machine learning and AI methods to create data-driven intelligent, non-fragile systems for automation, augmentation and amplifications. MI is about the ability to augment human intelligence. Humans will be empowered by an ecosystem of sharing data and insights, with support from digital assistants guiding and augmenting human awareness. MI will create a new type of autonomous coaching environment where humans and machines can train and mentor each other. This situation is comparable to a teacher in a classroom guiding, mentoring, discussing with and learning from the students.

MI will help us understand and accomplish things we never would have discovered by ourselves. It establishes a whole new foundation and potential for innovation, with the ability to be much more influential than industrialization.

Human-machine communication will further evolve toward a multifaceted communication platform that includes capabilities such as situation and social awareness. A deeper dialogue between humans and machines will emerge, moving beyond cognitive intelligence toward augmented human intelligence.

SIMPLICITY THROUGH ANALYTICS AND AUTOMATION

Among the many tools in the MI toolbox is analytics – covering everything from straightforward analytics over multiple nodes and petabytes of data, to complex multidimensional analytics on parallel processor systems. Today, analytics is often a human-involved process that requires the consideration of several aspects, including how to handle data volumes, data speed and the multitude of data types.

An example of complex event process handling with a massive amount of real-time data is electronic trading. This type of tool is capable of handling, analyzing and drawing conclusions based upon millions of messages per second. From a network perspective, we will see these types of use cases further substantiated by the evolution of the IoT. Connected cars and other types of connected devices will come online, providing more use cases that require real-time messaging from a variety of decentralized data sources.

Jettison and metadata are essential to handle the huge variety in data structuring. Automating these tasks is of great

importance to any enterprise with digital aspirations because human-intense interventions are not scalable.

Automation is best described as a closed-loop system. The automation closed loop refines system intervention depending on recorded impact, with minimal latency. The intervention is updated based on feedback on the system performance. The closed loop introduces the necessary changes without human intervention, based on performance goals defined by humans.

We are entering the era of early enablement of sentient MI that can be used to create digital attention, agile memory and goal management.

NETWORK EVOLUTION BRINGS MUCH MORE THAN RAW DATA

Over the next decade, industry and society will establish a foundation of insight-driven systems leveraging MI technologies. New service engagements will emerge, driven by predictive modelling and automated operations. Furthermore, a variety of deployment models serving different use cases will emerge, such as pure cloud-native applications, on-premises data center operations, and distributed deployments across multiple sites. All of this will be beneficial to many digital businesses and industries.

Analytics, MI and automation capabilities will be an integral part of future networks, substantiating innovation from network operation to new business opportunities within the IoT, for instance. ☺

#3 END-TO-END SECURITY AND IDENTITY FOR THE IOT

WHILE THE IoT is undoubtedly full of promise, there are still concerns about the proper handling of security and privacy, especially within mission-critical industries. Cybersecurity threats are emerging rapidly at the same time as the volume of the connected devices and software is increasing. It is more essential than ever to take a comprehensive approach to security and privacy that ranges from devices and gateways with connectivity to the cloud, IoT platforms and applications; chips to services; and development to operations. The need for industry-wide and cross-industry collaboration must also be taken into account.

This comprehensive approach includes end-to-end management of security and privacy that provides predictive security insights and, in many cases, automated adjustments based on policies. Security and privacy are not things that can be added on; they must be fully integrated into domains and components, processes, storage and communication.

A SYSTEM BUILT AND ENABLED BY TRUST COMPONENTS

The comprehensive approach to security and privacy will be based on an industry-wide agreement on how to secure trust in development, deployment and operations. It will include security and privacy enabling development tools, along with tools for automated and secure deployment. It will also include operations that provide direct

feedback to development, continuous compliance monitoring, and in many cases, automated, policy-based security and privacy orchestration that considers the constantly changing threat, vulnerability and trust landscapes.

In the coming years, identity management technologies will continue to be in the spotlight – along with technologies that enable trust in business-critical data and privacy-related personal data. Blockchain is an emerging technology that has potential in this regard. Security analytics and machine learning technologies will provide security insights about threats, vulnerabilities and security status. System integrity will be based on root-of-trust technologies that enable trustworthy hardware and software components.

The trust-enabling security and privacy technologies implemented in devices should be cost-effective and highly scalable. Mission-critical industries in particular will require high trust in security and privacy, as well as the ability to meet tight time constraints.

A COLLABORATIVE INDUSTRY EFFORT

As a part of the comprehensive security and privacy approach, collaboration in threat, vulnerability and trust exchange will increase. Different industry players can contribute to the collaboration within their area of expertise.

Greater transparency will boost trust, which will in turn boost IoT adoption and accelerate digitization in mission-critical industries. The comprehensive approach to security and privacy together with industry-wide collaboration, joint trusted development and standardization will be essential trust enablers. Since network service providers are ranked among the most trusted industry players (according to sources such as the 2015 Accenture Digital Consumer Survey), Ericsson is committed to helping them play this key, trust-building role across multiple industries. ☘

#4 AN EXTENDED DISTRIBUTED IOT PLATFORM

THE TECHNOLOGY shifts that the IoT brings will transform the technology industry in unprecedented ways. Networks will facilitate transactions, operations, logistics and the like, and simultaneously collect and analyze data, enabled by a cloud-based network infrastructure that connects machines, vehicles and devices. The race is on to develop and expand the capabilities of an industry-wide IoT platform. New business opportunities will be born through cross industry-society engagements.

An industry-wide IoT platform will be made up of decentralized devices ranging from simple passives to autonomous devices that are connected and communicating through a distributed cloud with a horizontal application and management platform. It is a distributed system, where insights are aggregated from the edge toward the center and regulated by policy and security settings that are specific for each application and use case. This is a multi-cloud based infrastructure, where a combination of public and on-premise solutions connect and serve any type of IoT-related service from near-product to software-as-a-service.

The Industrial IoT (IIoT), a branch of the IoT that is optimized for industry-specific use cases with enterprise connectivity, is also emerging. Production flexibility is enabled by system functionalities that have both specific nomadic and mobility capabilities. Everything centers around different aspects of logistics – not only from a physical perspective but also around information flow and data streams. By setting policy roles, network slices are created, defining what is allowed for each specific use case. Secure handling of information is essential, so that data can neither leak in or out of the system.

SEMANTIC INTEROPERABILITY TO SECURE FUNCTIONALITY GROWTH

Devices come with different monitoring, management and security functions. Robots in warehouses, for instance, require position tracking and coordination and delivery functions. Simple home automation solutions use cloud-based applications that are centrally connected to other information flows, such as weather forecasts, traffic cameras and so on. A connected vehicle cloud includes functions such as infotainment, over the air upgrades, telematics, remote

control, vehicle safety and security, fleet management and emergency services.

Pre-provisioned devices are one of the building blocks that enable an IoT platform that can grow in functionality as new device types connect to the network. The first time these devices “wake up” and connect, they inform the system who they are, what they can be used for and what they are capable of. The system is further enhanced by zero touch provisioning, including fully automated device cycle management.

Devices become an integral part of the future network and incorporate connectivity among clusters of devices through a combination of wireless and wired access technologies. The wireless connections are based on both wide area and short range technologies.

SEMANTIC APPLICATIONS AND SERVICE SOLUTIONS

The extended distributed IoT platform brings major opportunities to the business landscape when enabling a wide range of new digital services. It's also a model that moves from traditional heavy capital investments toward an operational-expenses-centric model. This is manifested through an adoption

and evolution of various types of “as a service” models.

A full palette of new flexibility is brought into the market as pay-as-you-go/use/grow enables SMEs to offer far better services than they do today. These services include proactive and predictive maintenance, where the service improves over time as a result of the ability to learn from data.

Ericsson's IoT-related engagements range end to end. For example, we are examining how access technologies can be optimized for various types of IoT use cases such as NB-IoT (NarrowBand IoT) to reduce power consumption and cost. Ericsson is also investing heavily in the evolution of LTE and the enablement of 5G networks to extend the range of addressable IoT use cases and

applications. Additionally, we are working to create cross-industry engagements such as 5GAA to connect the automotive and telecom industries (including devices) to develop end-to-end solutions for future mobility and transport services. It is vital for a healthy, evolving market to have industry-wide standards that reduce fragmentation, which would otherwise hamper IoT market adoption. *

#5 OVERLAYING REALITY WITH KNOWLEDGE

IN THE LAST couple of years, many devices capable of rendering immersive experiences have reached the consumer market. The virtual reality (VR) market has mainly been driven by the gaming industry but the technology has now been picked up by studios and other content creators. Augmented reality (AR), which offers a user experience that connects to the physical location, is also becoming popular. Real-time information is overlaid on whatever the user looks at, which enables the user to understand reality in greater depth.

AUGMENTED KNOWLEDGE VIRTUALLY EVERYWHERE

One example of AR is a task-based digital assistant that simplifies complex tasks with real-time guidance. This type of system is able to provide feedback as soon as a mistake is made by an operator or field service engineer. The feedback helps the practitioner become more knowledgeable and take immediate action to fix the mistake.

Another promising use case combines VR and AR services. In this scenario, a user is assisted by a remote technician to solve a difficult problem. The user shares a real-time video of the surroundings with the remote technician. The technician analyzes the video and instructs the user

on how to solve the problem by sending audio and overlaid graphics to the user's AR glasses.

Within the area of AR, a wide range of use cases emerge besides remote mentoring. Many practical applications are also recognized in digital industrial workplaces, which are sensor-rich environments with networked machine and computational power available for analytics of sensor and machine data. Objects that are currently offline can easily be connected and augmented through computer vision technologies. This is an extension of immersive applications in a typical IIoT scenario that provides increased productivity through improved uptime, quality and safety. Additional support tools within IIoT include visualization of data, document navigation and employee training.

PUSHING TOWARD REALITY PARITY

The technology evolution for immersive solutions requires a wide range of tools and infrastructure. These tools include display technology, real-time eye tracking, volumetric capture, perceptual computing for locations and surrounding positions, body movement and more. High-resolution cameras, microphones, GPS, gyros, connectivity, battery, voice and gesture control are also examples of components included in the concept. Compared with today's smartphones,

the obvious difference is within the man-machine interaction.

Over the next decade, computer vision will get better through 3D mapping, improved field of view, full-color depth and holographic technologies. With improved compute capabilities comes reduced disturbances from latency and rendering. VR is expected to reach parity with reality and thereby enable true 3D communication.

VR and AR are compelling use cases for 5G because they require high data rates and low latency. Change of viewport when turning the head requires low latency or the user will suffer vertigo. Motion to photon delay should be less than 20ms. End-to-end delay and latency requirements are key to providing a pleasant user experience. Bandwidth may become high (for the uplink video) depending on the use case.

Unfortunately, the current VR/AR market is fragmented with many verticals for different cameras, workflows and headsets. To address this issue, Ericsson became a founding member of the VR Industry Forum, which creates guidelines and interoperability along the entire end-to-end chain. The scope is to further the widespread availability of high-quality audiovisual VR and AR experiences for the benefit of consumers. 📺

Tackling IoT complexity

WITH MACHINE INTELLIGENCE

Internet of Things (IoT) applications transcend traditional telecom to include enterprise verticals such as transportation, healthcare, agriculture, energy and utilities. Given the vast number of devices and heterogeneity of the applications, both ICT infrastructure and IoT application providers face unprecedented complexity challenges in terms of volume, privacy, interoperability and intelligence. Cognitive automation will be crucial to overcoming the intelligence challenge.

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The IoT is built on the concept of cross-domain interactions between machines that can communicate with each other without human involvement. These interactions generate a vast number of heterogeneous data streams full of information that must be analyzed, combined and acted on.

■ To be successful, providers of ICT infrastructure and IoT applications need to overcome interlinking challenges relating to volume, privacy, interoperability and intelligence. Doing so requires a multifaceted approach involving concepts and techniques drawn from many disciplines, as illustrated in *Figure 1*. Research in 5G radio, network virtualization and

distributed cloud computing primarily addresses the volume challenge, by evolving network infrastructure and application architecture design to increase the amount of resources available to applications (throughput, compute and store resources, for example). Meanwhile, the privacy and interoperability challenges are being addressed with a mix of R&D efforts and standardization activities that are leading to novel concepts and techniques, such as differential privacy, k-anonymity algorithms, secure multiparty computation, ontology matching, intelligent service discovery, and context-aware middleware layers. Addressing the intelligence challenge requires the development and use of intelligent decision

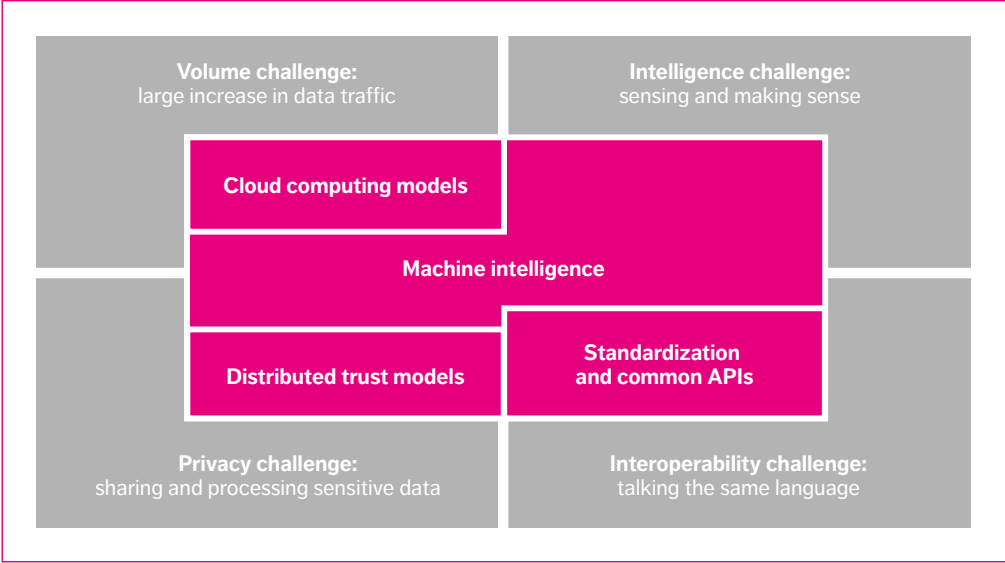


Figure 1
Key challenges (in gray) and approaches to addressing them (in pink)

support systems (DSSs) – interactive computer systems that use a combination of artificial intelligence and machine learning methods known as machine intelligence (MI) to assist and enhance human decision making. In response to this need, Ericsson has built a cognitive automation framework to support development and operation of intelligent DSSs for large-scale IoT-based systems. The use cases are not only the traditional telecom ones (such as operations support systems/business support systems automation), but new ones in domains such as transportation, robotics, energy and utilities, as well.

The main benefit of our cognitive automation framework is that it reduces DSS development and deployment time by reusing as much knowledge as possible (such as domain models, behaviors, architectural patterns, and reasoning mechanisms). It also cuts operational costs for IoT-based system management by making it possible during runtime for a DSS to decide automatically how to adapt to changes in its context and environment, with minimal or no human interaction.

While DSSs predominantly address the intelligence challenge, they can also be used to address privacy, volume and interoperability challenges. For example, IoT devices need intelligence to learn to trust each other. Management of network infrastructure may also benefit from intelligent systems, to handle the large volumes of transmitted data more efficiently. Finally, systems that use different standards and/or APIs can use MI to interface with each other, by using language games, for example.

Architectural components and structure of the knowledge base

A high-level conceptual view of the architecture of our framework is depicted in *Figure 2*. The main components are the observer, the knowledge base, the reasoner and the interpreter. The observer is responsible for pushing knowledge from the environment to the knowledge base. This is done by first deriving a symbolic representation of data using model transformation

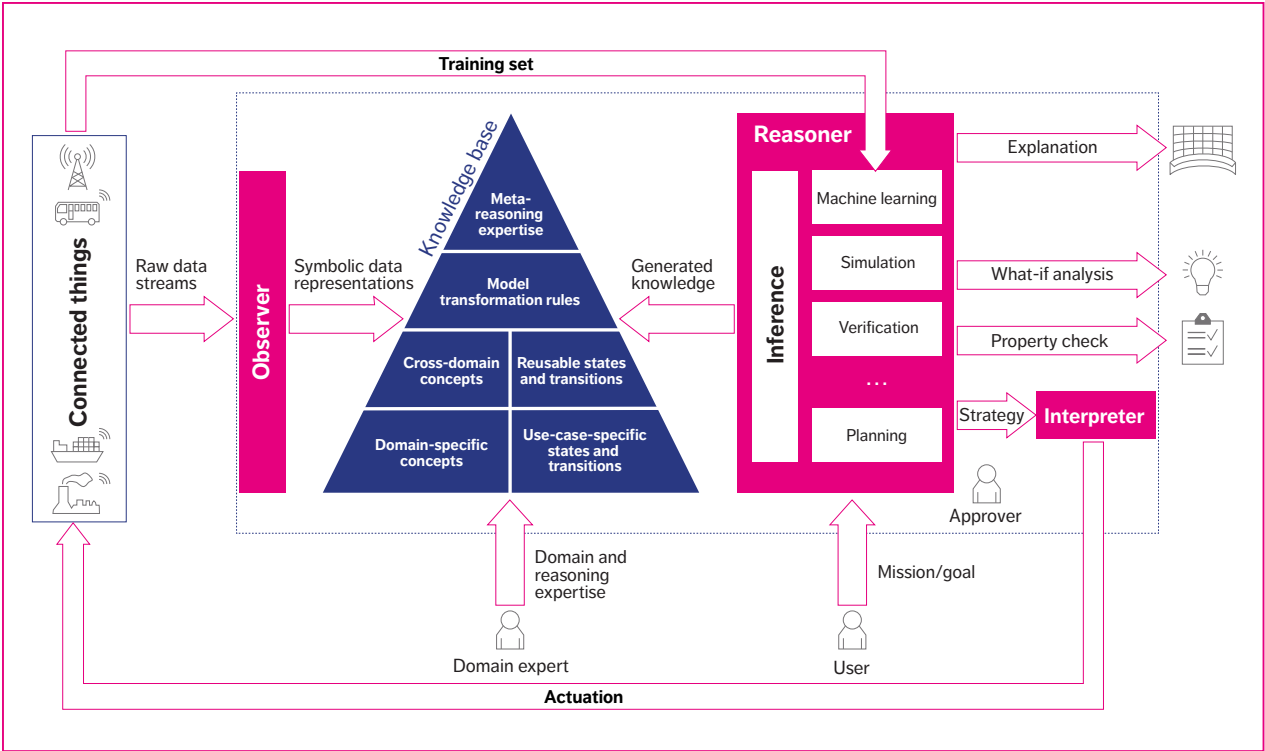


Figure 2
A high-level, conceptual view of the framework architecture

rules that identify the relation between raw data coming from the IoT-based system and a symbolic representation of the captured data. The next step is to perform ETL (extract, transform, load), followed by a semantic analysis of the obtained data.

The knowledge base contains a formalized description of general concepts that can be used across domains to facilitate interoperability, as well as domain-specific concepts (transportation, for example). In addition, the knowledge base contains the possible discrete states of the system, potential transitions between states, meta-reasoning expertise and model transformation rules. Ontologies in RDF/OWL semantic markup language are among the possible formats of the knowledge stored in the knowledge base [1].

Domain experts can enter domain and reasoning expertise directly into the knowledge base. Since the behavior of the environment is dynamic and unpredictable, the knowledge base is continuously updated with knowledge coming from the observer and the reasoner.

Most of the framework intelligence comes from the reasoner, which contains general purpose inference mechanisms that allow it to draw conclusions from information (propositions, rules, and so on) stored in the knowledge base, and problem-specific tools that implement various MI tasks such as machine learning, planning, verification, simulation, and so on. By relating a goal/mission query to meta-reasoning expertise, the inference engine selects an appropriate method or problem solver, and derives

relevant input for it using model transformation rules. The result produced by the selected problem solver can be presented to the user, sent to the interpreter for actuation, or fed back into the inference engine to reinforce the knowledge base content.

A goal query can be initiated by the user or the inference engine itself based on a comparison between expected and actual system states. For instance, if the goal query is to verify a certain system property (safety, for example), the inference engine will look in the knowledge base and deduce that a verification method should be used to check that the system satisfies the required property. Similarly, if the goal query is to reach a certain goal, the inference engine will look up the knowledge base and deduce that a planner should be used to generate a strategy to reach that goal. Since most of the planners accept Planning Domain Definition Language (PDDL) [2] as their input, the inference engine looks up corresponding model transformation rules to translate the problem in PDDL. The interpreter takes the generated strategy and transforms it into actuation instructions for the connected things. When executing the strategy, the interpreter works closely with the reasoner. In the event of any changes in the expected state of the system, the reasoner sends a replanning request to the planner. The generated strategy can be presented to the user for approval before actuation, if required.

The reasoner can perform automatic knowledge acquisition using machine learning techniques that may extract insights (such as categorization, relations and weights) from training sets in the form of documentation, databases, conversation transcriptions or images. The reasoner can explain the reasons behind recommending and performing a given set of actions, trace back to the origins of the decisions, and inform the user (and other systems, if so desired) about actions planned for the future. The explanation can be made in a user-friendly manner by applying natural language processing, augmented reality and automatic speech synthesis and recognition, for example.

THE REASONER CAN PERFORM AUTOMATIC KNOWLEDGE ACQUISITION USING MACHINE LEARNING TECHNIQUES SUCH AS CATEGORIZATION, RELATIONS AND WEIGHTS

Application examples
Our research and experience show that there are a wide variety of ways in which our framework can be used to boost efficiency in different types of applications. Test Automation as a Service (TAaaS), automated scheduling of train logistics services, smart metering, ticket book systems and Everything as a Service (XaaS) are just a few examples.

TAaaS
Our TAaaS project addressed the product development life cycle and software-testing activities. Testing for products typically involves numerous tools for test design, test execution and test reporting, along with management software tools, put together in a toolchain. Toolchain configuration is an expensive process that often takes one to two days to prepare, depending on the product being tested. Reconfiguring a toolchain to test another product is an even more time-consuming activity. The TAaaS system automated the configuration of these toolchains based on user requirements and created virtual workspaces in a data center, eliminating the need for expensive, manual configuration. In this case, the framework contained a knowledge base of software tools and their dependencies.

Automated scheduling of train logistics services

In this project, we created a concept for a fully automated logistics transportation system. The system consisted of several actors (such as trains, loading cranes and railroad infrastructure elements) equipped with sensors and actuators, and coordinated by intelligent software. The main goal was to investigate the potential benefits of combining factual and behavioral knowledge to raise the level of abstraction in user interaction. A user-specified, high-level objective such as “deliver cargo A to point B in time C” is automatically broken down to subgoals by the intelligent management system, which then creates and executes an optimal strategy to fulfill that objective through its ability to control corresponding connected devices.

Smart metering

We used our framework to create an intelligent assistant to help Ericsson’s field engineers manage Estonia’s network of connected smart electricity meters. Its task was to automate troubleshooting and repair procedures by utilizing knowledge captured from the domain experts. By separating knowledge from the control logic, we made the system extendable so that new knowledge can be added to it over time. The representation comprises several decision trees

CONSUMER SERVICES CAN BE DELIVERED AUTOMATICALLY USING UNDERLYING MODULAR COMPONENTS KNOWN AS MICROSERVICES THAT CAN BE REUSED ACROSS DIFFERENT APPLICATION DOMAINS

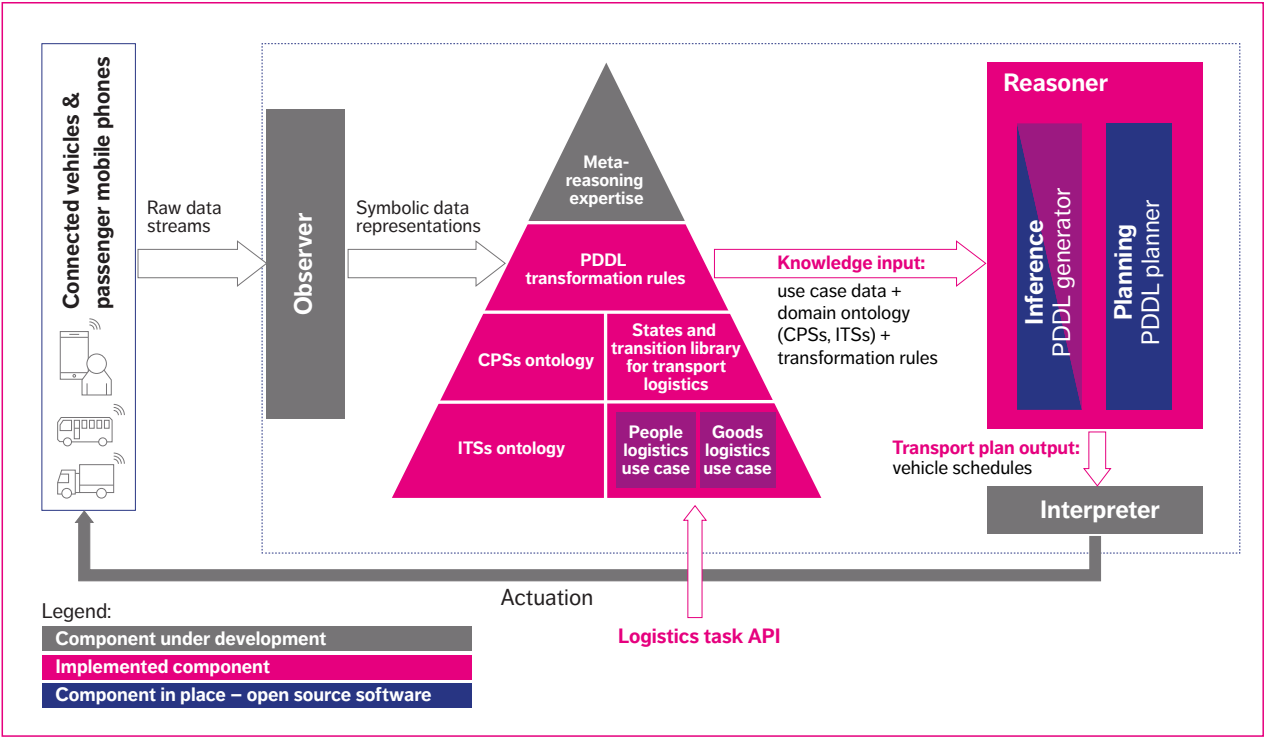
and workflows that encode the root causes of various potential technical issues, processes for diagnosing and identifying them, and the steps in the corresponding repair procedure. Our project revealed that in this type of use case, the bulk of the work is in the manual acquisition and encoding of expert knowledge.

Ticket book

When managing and operating mobile networks, support engineers normally create tickets (issue descriptions) for problems classified as non-trivial. Based on the assumption that the solution to an issue similar to another that has previously been dealt with will be similar, we created a system for semantic indexing and searching historical tickets. We used a vector space model to calculate a similarity score and sorted searched tickets according to their relevance to the current case. A similar technique was used to find technical documentation relevant to the case currently being solved. This use case relies on the availability of historical data (previous tickets) to automatically build a knowledge base that facilitates subsequent information retrieval and relevancy ranking.

XaaS

In our XaaS project, we developed an MI-assisted platform for service life cycle management. By leveraging technologies such as semantic web and automated planning, we presented how consumer services can be delivered automatically using underlying modular components known as microservices that can be reused across different application domains. This automation provides flexibility in service deployment and decommissioning, and reduces deployment time from months and weeks to minutes and seconds. The knowledge base comprised a specification of the domain used for service requirement formulation, predefined service definitions, the metadata of the service functional components, and the descriptions of the microservices that had been used to implement those components.



Prototype implementation – transportation planning

The automation of a transportation planning process is an excellent example of how our cognitive automation framework can be used to create an intelligent DSS. In this example, the DSS is used to automatically determine how to transport passengers or cargo at minimal cost (taking into account the distance traveled, the time needed to transport each passenger and piece of cargo, or the number of buses required for transportation) using connected vehicles (buses or trucks respectively). The answer for a particular task could be a plan that consists of a sequence of steps for the system to perform to reach the goal state. If the system determines that the task cannot be performed, the answer from the framework-based DSS could be the reason why,

as well as a possible solution such as increasing the number of vehicles. The motivation behind using the framework for this particular use case was to demonstrate the decrease in the cost of design by building and reusing cross-domain knowledge specifically between people and cargo logistics.

Figure 3 provides an overview of the partially implemented prototype implementation of our framework-based DSS for transportation planning. (Work is still ongoing to implement missing components connecting the framework to the environment.) The knowledge base contains model transformation rules for PDDL and states and transition models that are based on a set of ontologies containing information for the particular transportation domain. The ontologies are organized in multiple layers of abstraction: one

Figure 3 Overview of partially implemented framework-based DSS for transportation planning

DOMAIN KNOWLEDGE	DESCRIPTION
Route	Specification of route(s) as graph(s), which includes vertices, edges and edge-traversal costs (such as travel time and fuel spent).
Transport agents	Number of vehicles, vehicle IDs and vehicle capacity.
Transportable entities	Number and ID of transportable entities (passengers, cargo).
Initial state	The starting location of the transport agents and transportable entities.
Transitions	The transitions that transport agents can perform. Current prototype implementation contains three transitions: pickup, drop, and move-to-next-coordinate.
Goal state	Which criteria need to remain constant for the transport service to complete the specified route (usually this means that all transportable entities are picked up and dropped off at specific points along the route).

Figure 4
Knowledge for
transport plan
generation

which is cross-domain (the cyber-physical systems ontology in Figure 3); and one which is specific to intelligent transportation systems (ITSs), used for reasoning about transportation problems.

This implementation assumes that the inference engine has already deduced that a planner should be used to solve the transportation planning problem using meta-reasoning expertise (from the knowledge base) and a user-specified goal. The other implemented component is the PDDL generator, which – given the PDDL transformation rules, states and transition files as input – generates files that are understandable for a PDDL planner. The source code of the implementation is available online [3].

As always, there is a direct relationship between the amount of knowledge to be formalized and the effort required to formalize it. Put simply, this means that the more non-formalized knowledge there is, the higher the operational costs will be in terms of time, human resource allocation

and money. In the case of a transport schedule generation process, the benefit of using the framework to lower the cost of design is that it reduces the effort required to formalize the knowledge needed for the system to be automated.

Figure 4 shows the different types of knowledge used to generate a transportation plan for two use cases: transportation of passengers in buses and transportation of cargo in trucks. The transport network is viewed as a graph, with points of interest (POIs) as vertices and POI-connecting roads as edges. The “transportable entities” can be passengers or cargo, depending on the use case. Transitions are similar regardless of the route, number and type of transport agents (buses or trucks) and transportable entities. This means that reusing transitions across different transport planning use cases, and storing them as part of the “transition library” (see Figure 3), can reduce the cost of design.

THE FRAMEWORK'S SELF-ADAPTATION IS SUPPORTED BY META-REASONING AND CONTINUOUS REINFORCEMENT OF THE INTERNAL KNOWLEDGE OVER TIME

We use a simple metric called the reusability index to measure reductions in the cost of design. The reusability index is the ratio of reused entities versus total entities in the knowledge input available to the PDDL generator to generate the plan. The ratio was 0.364 for the bus case and 0.251 for the truck use case. This means that out of the total number of entities created for each case, 36.4 percent were already available in the library for the bus case and 25.1 percent for the truck case. The contrast between the two can mainly be attributed to the difference of the route specification entities, as agents on both routes followed different paths. As the knowledge base becomes more populated, the reusability index will increase, which will in turn lead to a decrease in the development time of further custom ITS solutions. Our measurements also indicate that the reusability index can rise if the route specification is part of the PDDL generator, which can automatically generate the specification using data from a mapping service in conjunction

with a routing library. The knowledge engineer could then only specify the desired waypoints (bus stops, for example), and the routes and graphs would be generated automatically by the PDDL generator.

Conclusion

IoT-based systems require a large number of decisions to be made in a short time, which in turn requires automation – not only in terms of infrastructure management, but also within the logic of the IoT applications themselves. A DSS is an essential tool in this context, owing to its ability to enhance human decision-making processes with MI based on behavioral models and information contained in the relevant data streams.

Any system that requires automation in the decision-support process could be enhanced by our cognitive automation framework. It has a domain-agnostic, fixed architecture in which the only variable components are the knowledge base and the set of reasoning methods. By separating cross-domain knowledge from use-case-specific knowledge, the framework makes it possible to maximize reuse, enabling substantial savings in terms of design cost, and shortening time to market for custom-developed solutions. The set of reasoning methods is extensible and can be populated with the methods and tools that are best suited to specific tasks. The framework’s self-adaptation is supported by meta-reasoning and continuous reinforcement of the internal knowledge over time. In the future, we plan to extend the framework with analysis of hypothetical situations and an intuitive user interface for both experts and less experienced users. *

Terms and abbreviations

API – application programming interface | CPS – cyber-physical system | DSS – decision support system | ETL – extract, transform, load | IoT – Internet of Things | ITS – intelligent transportation system | MI – machine intelligence | OWL – Web Ontology Language | PDDL – Planning Domain Definition Language | POI – point of interest | RDF – Resource Description Framework | TAaaS – Test Automation as a Service | XaaS – Everything as a Service

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Further reading

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- » **The Networked Society will not work without an automation framework, Ericsson Networked Society Blog (July 2015), available at:** <https://www.ericsson.com/thinkingahead/the-networked-society-blog/2015/07/09/the-networked-society-will-not-work-without-an-automation-framework/>

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DevOps: Fueling the evolution.

TOWARD 5G NETWORKS

DevOps has an important role to play in meeting 5G networks' requirements for faster time to customer in an environment characterized by widely distributed resources and tight constraints on service quality. In collaboration with the open source and academic communities, we have investigated how best to address 5G challenges using DevOps and a generic architecture focused on agility and flexibility.

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DevOps approaches extend the agile software development culture to deployment and operations, balancing the development team's desire for rapid change with the operations team's desire for stability.

■ In enterprise environments, DevOps processes and techniques that rely heavily on automation are credited with enabling significant increases in the efficiency of the software delivery cycle all the way into operations. As part of the transition to 5G networks, telecom vendors and operators alike

are considering how to adapt DevOps ways of working to boost competitiveness by shortening feature delivery cycles and raising feature hit rates through feedback loops.

5G is expected to deliver unprecedented performance in terms of transmission capacity and packet transit delays, enabling new applications and services in areas as diverse as the Internet of Things, augmented reality and the Industrial Internet [1]. To dynamically define the features supported by the infrastructure and the ways in which these features are managed, 5G networks

will rely heavily on virtualization technologies [2], as shown in *Figure 1*.

Network Functions Virtualization (NFV) [2] plays a key role. NFV disaggregates the network function (for example, the router, mobile packet gateway, firewall) from the physical box that contained it. This enables its software implementation to be optimized for deployment on a distributed cloud infrastructure, where an appropriate set of resources may be provisioned dynamically to control resource utilization, energy consumption, and coverage, for example.

DevOps in next-generation telecom networks

The evolution toward virtualization transforms the way both equipment vendors and telecom operators work. *Figure 2* illustrates how DevOps can be used to optimize a software delivery cycle, including everything from feature development to operations across disciplines (development, customer engagement and operations) through to continuous delivery (CD) practices. Focusing on automation and lean management practices enables flow control and transparency across the cycle. The organizational and administrative interfaces between the different actors within

CD PRACTICES AIM TO OPTIMIZE THE FLOW OF SOFTWARE THROUGH THE SOFTWARE DELIVERY CYCLE

the 5G ecosystem must be easy to traverse, with appropriate software to secure continuous automation flows.

In a 5G context, the word software refers to both the actual code of virtual network functions (VNFs) and models describing the infrastructure and execution environments hosting this code. While software flows clockwise through the cycle, each stage provides feedback to the previous one (counterclockwise) to allow for software quality improvement and process optimization.

CD practices aim to optimize the flow of software through the software delivery cycle. Through comprehensive, fast and reliable test and deployment automation, it is possible to achieve higher release and deployment frequencies. This leads to shorter time to market and time to customer,

DevOps and continuous everything

DevOps is an interactive approach to product management, development, deployment and operation that stresses communication, collaboration, integration and automation. Working together with the customer every step of the way, the DevOps approach begins with requirement setting and continues through development and operations.

Continuous integration – Automated process of secure and frequent integration of source code into source baselines, and binaries into system baselines.

Continuous delivery – Automated process of secure and frequent internal provisioning of ready-to-install software product versions of integrated software.

Continuous release – Automated process of secure and frequent provisioning of delivered software product to external customers and clients.

Continuous deployment – Automated process of secure and frequent production, testing and/or monitoring, and deployment of software products to customer equipment in a live environment.

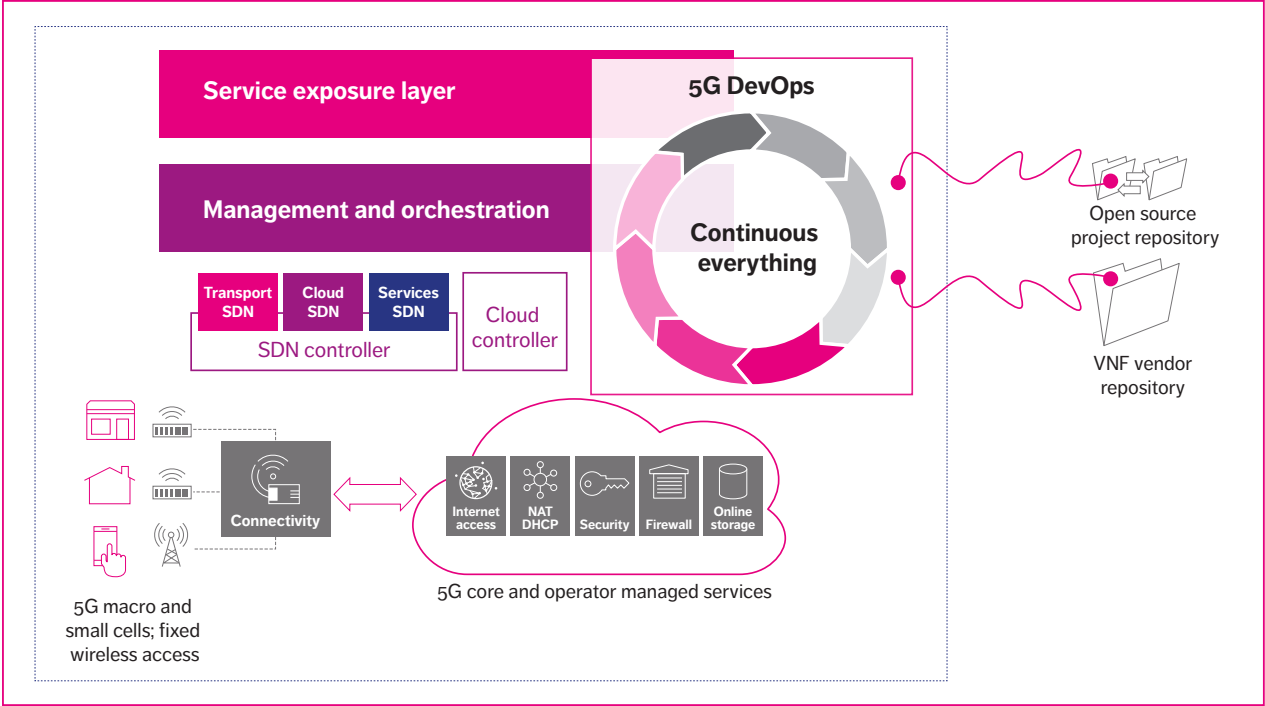


Figure 1 5G network and function ecosystem

and it enables improved responsiveness to customer and market demands.

Maintaining one track in software development, using feature flag-driven development, and establishing version-controlled repositories for application code and application and system configuration data enables teams to create a complete environment that is ready for consistent “build and deploy”. Lean management practices aim at process improvement through effective work in process limitations, the monitoring of quality and productivity, as well as the use of application and infrastructure monitoring tools as part of the feedback loop to steer development.

Both CD and lean management practices tie together continuous everything (continuous integration, delivery, release and deployment)

activities across teams and stakeholders.

When implementing these practices, a number of methods and tools are used on top of an architecture, which provides the capabilities for automation and transparency.

The architecture plays a significant role in building, deploying and operating complex systems. In NFV, it describes how high-level functions typically developed by different teams or open source projects can be interconnected and packaged together to provide a service. Capabilities defined by the NFV MANO architecture [3] allow for dynamic configuration of parameters, dimensioning and scaling a service to reach a wanted set of performance indicators or policies.

The architecture also needs to provide the means for automated monitoring and

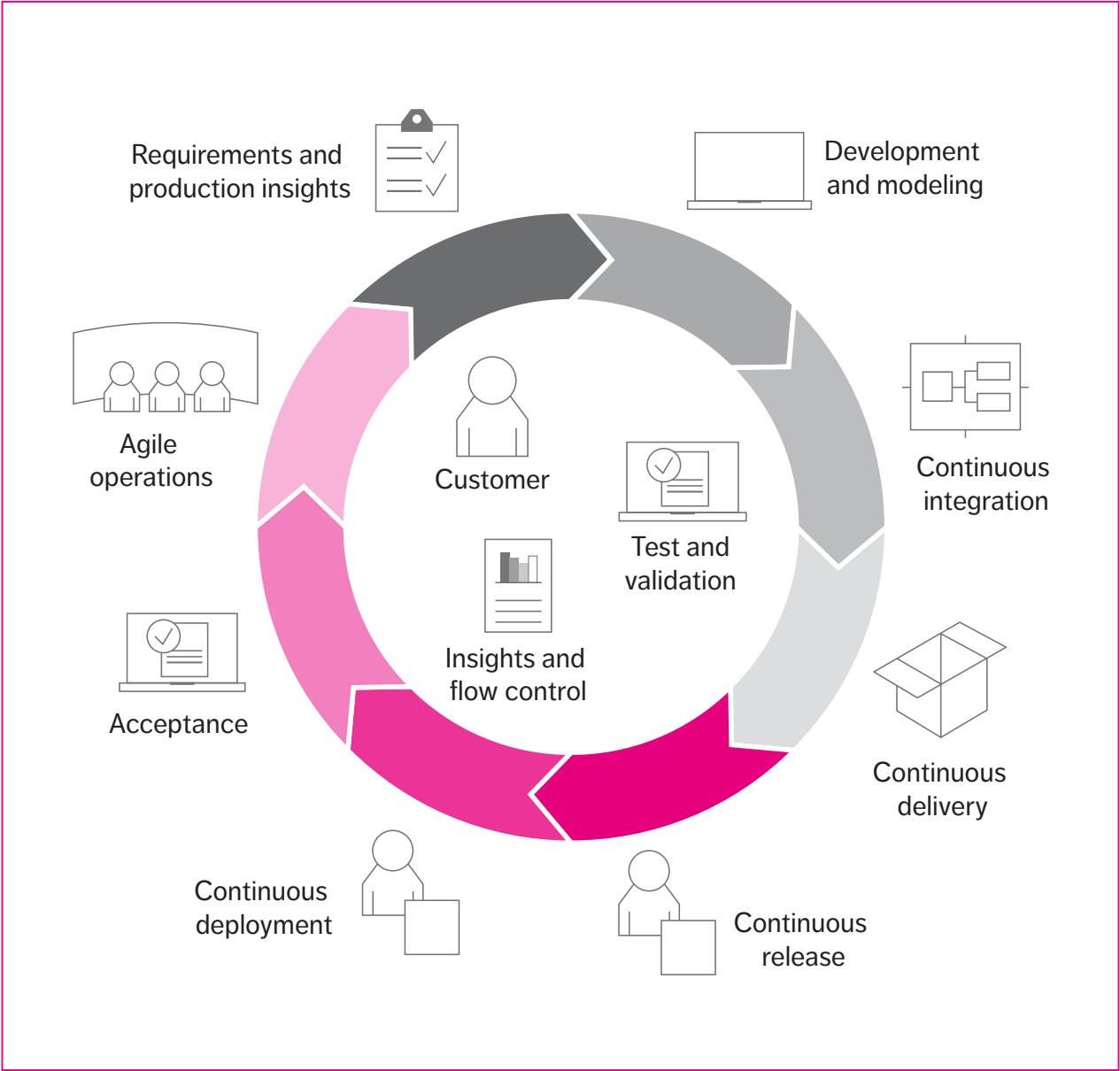


Figure 2 Simplified DevOps cycles

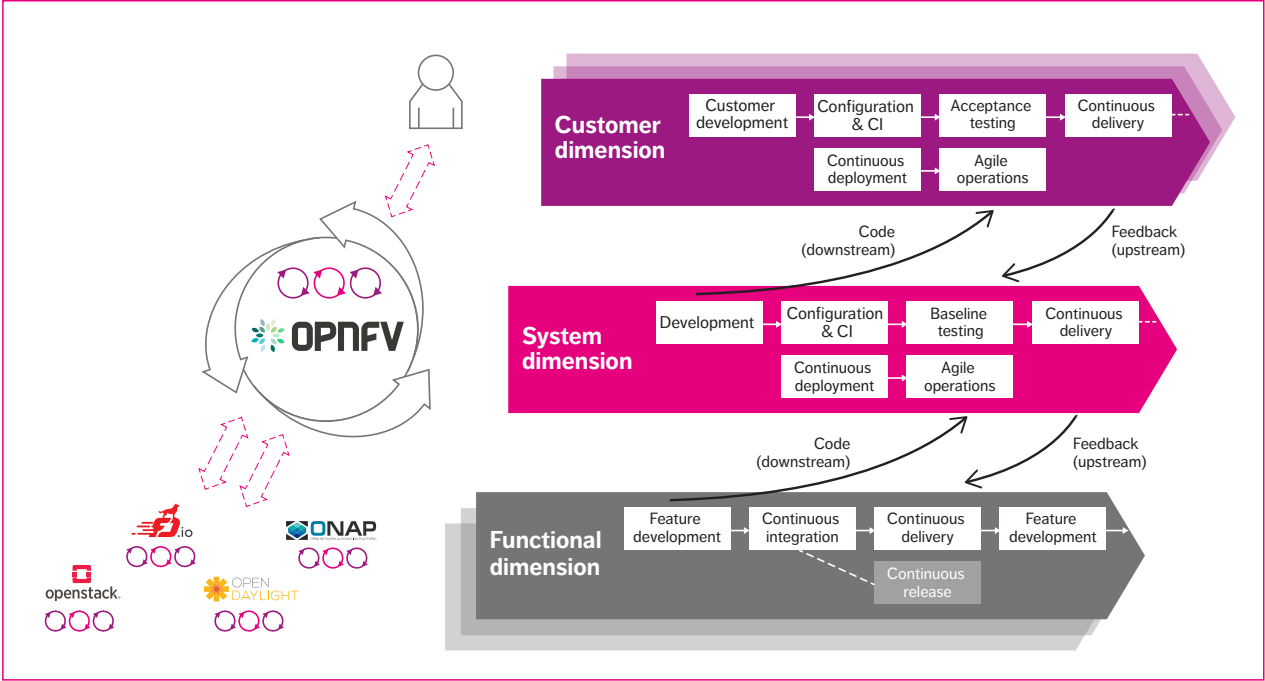


Figure 3 OPNFV and selected upstream projects mapped on DevOps dimensions

troubleshooting, so that advanced analytics can identify performance deviations from a wanted stage early on, and allow fault isolation in a large system, which eases resolution. Experiences and insights from implementing the automation, optimizing the software delivery cycle and operations are then used to drive the architecture management and improve development and testing of individual functions in a DevOps deployment.

Telecom-grade open source – a foundation for 5G

The emergence of NFV technologies has led to a significant increase in the number of open source projects specializing in different components of the NFV stack. A majority of them follow and apply continuous integration (CI) principles and practices to ensure that technical solutions can be developed faster, integrated with other projects and tested in a fully automated way, as well as enabling tailored feedback to developers and users.

However, many open source projects test the components they develop only within their own context without integrating components from other communities. This results in very limited or non-existent end-to-end testing, potentially introducing difficulties when these components are used in a different constellation at a later time.

The open source project Open Platform for NFV (OPNFV) addresses this issue by performing systems integration as an open community effort. Ericsson leads the CI/CD activities within OPNFV, coordinating efforts across different open source communities to ensure the different actors in the NFV ecosystem move toward a DevOps model. OPNFV consumes components of the NFV stack from different upstream projects, integrating and deploying them together, and testing them together in its CI (Figure 3). Like other open source projects, OPNFV applies CI practices strictly. OPNFV brings up and tests the NFV reference platform in a completely automated fashion with no manual intervention, aiming for faster, tailored feedback.

OPNFV strives not to keep any code for NFV components locally in its own source code repositories. When OPNFV identifies issues or

INCREASED DEVELOPMENT AGILITY AND FLEXIBILITY IN 5G WILL SUPPORT THE TRAFFIC GROWTH OF THE NETWORKED SOCIETY AND ENABLE NEW SERVICES

missing features, its developers propose blueprints or open bug reports to upstream projects that are then implemented directly in the upstream projects by the same developers. This is enabled by the different feedback loops OPNFV has established. Some of the open source projects OPNFV works with are OpenStack, OpenDaylight, FD.io and KVM.

Since it consumes and integrates components from upstream projects and tests the integrated platform, OPNFV can be defined as a downstream software project. Yet OPNFV also acts as an upstream software project by solving issues and implementing missing features directly in the upstream projects. The combination of the upstream and downstream behaviors therefore makes OPNFV a midstream project.

Due to its midstream nature, OPNFV faces a similar challenge to that of vendors and operators when it comes to integrating the components of the NFV stack to establish a working platform. In order for OPNFV to do the CI successfully, the upstream projects it consumes components from must do CD. Without it, OPNFV will have to wait for official releases rather than having early access to the latest stable versions of those projects. This would greatly limit the OPNFV value proposition by delaying the detection of faults in open source NFV components for months.

OPNFV Cross Community CI (XCI) aims to meet this challenge by providing a production-like environment to its upstream projects. By establishing

the feedback mechanisms between OPNFV and the upstream projects, open source communities are able to assess the maturity of their CI/CD journey, identify their own improvement areas and determine how they can contribute to making OPNFV successful. This experience is highly relevant for both vendors and operators that must integrate software from a variety of sources in 5G deployments.

Empowering developers in 5G

Increased development agility and flexibility in 5G will support the traffic growth of the Networked Society and enable new services. To illustrate this, we built a research prototype that targets the integration and delivery of distributed network functions for applications such as industrial robotics and media delivery, spanning multiple administrative and technology domains. We followed the fully automated DevOps pipeline in Figure 2, which addresses every step in the life cycle of an application.

We used common software development toolsets such as Git, GitLab, Advanced Package Tool and Jenkins to build an automated development and delivery pipeline. We integrated the pipeline with the network function life cycle orchestration by implementing software to facilitate the deployment from the development pipeline, life cycle management, policy control, and monitoring capabilities in addition to service and resource governance.

Our orchestration templates were based on several languages: USDL for the service modeling, TOSCA for network slice modeling and YANG for the resource modeling. To support a smooth flow of orchestration logic through the different abstraction layers, we developed transformation functions between the different models. An end-to-end-orchestrated industrial robotics application, shown in Figure 4, spans vertically through a business slice, a network slice, a system dimension and a functional dimension. Each slice/dimension has its own representation of the robotics application and its own abstraction of required resources. The application also spans horizontally across several distributed technology and administrative domains.

In the development and integration stage, after

SERVICE DEPLOYMENT ON 5G WILL LIKELY REQUIRE THE ORCHESTRATION OF A MULTITUDE OF TECHNOLOGY DOMAINS

initial testing, the code is tagged for packaging into the preferred library format. Libraries are stored in a dedicated repository, and in the integration stage, they are packaged together in the form of a binary software image such as a virtual machine or a Linux container.

Templates for service bundles and basic resource types are defined in the modeling stage. Templates can describe various aggregation levels, from very simple components such as a network function or network service to more complex product level components. For example, in the TOSCA case, templates also describe relationships, topology and life cycle workflows. Templates are stored in a blueprint repository. Multiple templates are aggregated to abstract product-related descriptions that refer to all dependent artifacts and customizable inputs.

Testing and validation are performed repeatedly, coupled tightly with the modeling stage, starting early in the cycle to eliminate errors and improve quality. As described in the TOSCA blueprints, we validate software components and their deployment, focusing on the aggregated types that represent building blocks of complex services. Validated blueprints and related artifacts are tagged as “ready for delivery” during the delivery stage and pushed to production repositories. Validated artifacts can be directly used for product offerings.

The deployment stage spans multiple orchestration levels for an automated end-to-end fulfilment of network services. Deployment artifacts are taken from the product repositories provided in the delivery stage. This flow starts with the uppermost business level, where customer requirements get mapped into the product offerings. Business level mapping is driven by the USDL service model and uses governance, pricing and resource negotiation inputs.

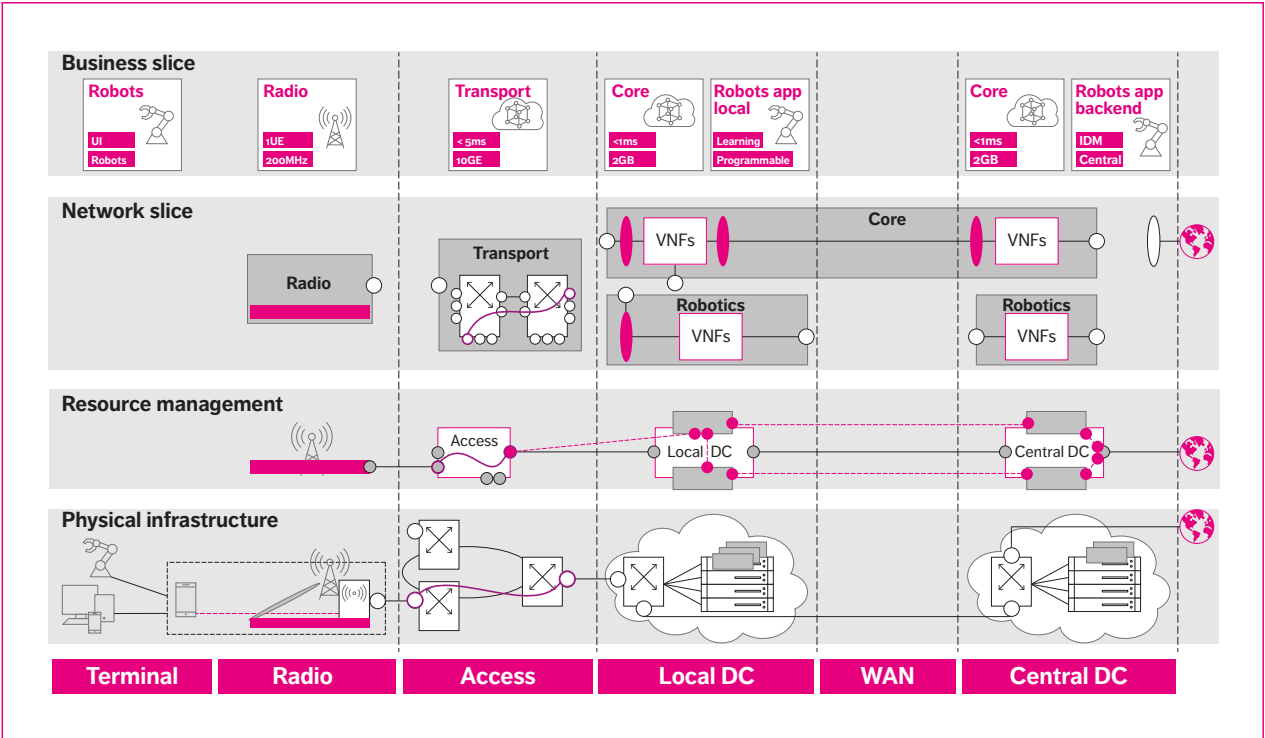


Figure 4 High-level modeling of 5G robotics application use case

Business service descriptions and Service Level Agreement requirements are further mapped to the network slice requirements and TOSCA blueprint deployment descriptions. The blueprints are customized with dimensioning values and deployment-specific parameters and pushed to the network orchestration layer. The blueprints are then processed by the life cycle manager, which drives the deployment workflow with available resources.

Service deployment on 5G will likely require the orchestration of a multitude of technology domains. Different domains contain specific resources with explicit capabilities and orchestration requirements. A hierarchical layering in the life cycle management makes it possible to cross the technology, functional and administrative domains. In such a hierarchical approach, life cycle management of the individual sub-domains is delegated to the lower orchestration layer. The upper layer handles end-to-end aggregation of sub-domains and higher abstraction-level life cycle management. Isolation properties are defined in the blueprint descriptions.

In the operations stage, a monitoring engine informs the policy engine of any deviation from the required quality level. The policy engine triggers life cycle workflows designed to maintain the quality of deployed services. Every orchestration layer contributes to maintaining overall service quality with its own optimized workflow.

Augmented operations capabilities

Together with 14 European academic and industry partners, we have designed and

developed a DevOps framework [4] for efficient deployment and operations of NFV-based services. The results addressed on-the-fly service verification, scalable and programmable observability, and automated troubleshooting.

An elastic router service is a good example of how augmented operations capabilities could be enabled throughout the DevOps pipeline [5]. The service is able to expand or reduce its capacity dynamically, in accordance with customer traffic demand. The elastic router is based on components openly available from upstream controller and virtual switch projects. It realizes elasticity by automatically scaling data plane resources as a function of traffic load.

The various stages in the DevOps life cycle outlined in Figure 2 continuously loop within the respective dimension shown in Figure 3 to provide rapid service agility and dynamicity. A natural entry point to the life cycle of a telecommunications service is the development of the functional service components, such as the elastic router components that provide dynamic scaling of forwarding functionality with centralized control. For this prototype, we used basic development tools such as local integrated development environments with Git/GitLab for code sharing, merging and versioning.

For integration and testing of the entire system (in other words, the complete elastic router VNF), we emulated the network scenario in Mininet, a realistic virtual network environment that is well established in the academic community. In a production system, this would be replaced by utilizing a dedicated

Terms and abbreviations

CD – continuous delivery | **CI** – continuous integration | **DevOps** – a compound of software development and operations | **GE** – Gigabit Ethernet | **Git** – version control system for tracking changes in computer files | **GitLab** – web-based Git repository manager | **IDM** – Internet Download Manager | **KVM** – Kernel-based Virtual Machine | **MHz** – megahertz | **NAT DHCP** – Network Address Translation Dynamic Host Configuration Protocol | **NFV** – Network Functions Virtualization | **OPNFV** – Open Platform for NFV | **SDN** – software-defined networking | **TOSCA** – Topology and Orchestration Specification for Cloud Applications | **UE** – user equipment | **UI** – user interface | **USDL** – Unified Service Description Language | **VNF** – virtual network function | **WAN** – wide area network | **XCI** – Cross Community CI

AGILE OPERATIONS
RELY ON TIGHT
INTERACTIONS BETWEEN
ANALYTICS AND
ORCHESTRATION
WORKFLOWS

network slice as a sandbox environment for staging, similar to the XCI as offered by OPNFV.

One or more production-ready system components (such as the elastic router VNF) are modeled as a directed graph before releasing and deploying the actual customer service. Such a joint model enables programmability of compute and network resources, making it possible both to define VNF placement and to establish the forwarding overlay in a single transaction.

By integrating new validation capabilities and related interfaces into the release and deployment phases of the customer dimension, service properties (modeled as graphs) can be verified before they are rolled out in the production infrastructure. This is essential in 5G and NFV environments, where reconfiguration of services must be triggered frequently. Misconfiguration of (V)NFs, violation of network policies, or artificial insertion of malicious network functions are a few examples of issues that formal verification methods can identify to ensure service uptime and preserve network integrity and reliability. In our research, we integrated new functions into the delivery, release and deployment components to support continuous, real-time verification of service programming instructions and configurations. These new functions act as gatekeepers, rejecting invalid models or configurations, and providing immediate feedback to the modeling stage. Validated service instances and configurations subsequently passed through the deployment and acceptance stages, which involved the automated mapping (or embedding) of the service model onto infrastructure

resources, and the introduction of the relevant service components to the infrastructure.

The agile operations stage starts once services and accompanying network configurations are successfully commissioned. Both services and individual service components must be observed continuously throughout their lifetimes. To support the requirements of future 5G networks, the project team developed and integrated a new software-defined monitoring framework that provides accurate and scalable monitoring both in large-scale, geographically distributed cloud scenarios and in centralized data center scenarios. The framework supports traditional and novel metrics together with local, lightweight analytics functionality, which trigger both local and remote reactions from the orchestration and management architecture in real time.

Agile operations rely on tight interactions between analytics and orchestration workflows. For example, local analytics performed on real-time monitoring data results in production insights, which can be used to support dynamic, autonomous service adaption (scaling) through the direct interface between the VNF and resource orchestration entities (such as Virtual Infrastructure Manager). Here, the service model would be automatically refined to accommodate the new scaling requirement by the VNF itself – that is, controlled by the VNF developer. The DevOps life cycle would remain in the customer dimension, reiterating the modeling stage in an updated service model.

Another example is a local analytics function that would identify problems related to the service functionality that cannot be solved by refining the service model. In this case, these production insights feed into our developer-friendly troubleshooting engine, performing root cause analysis with advanced automation support. Once the faulty service component is identified, the troubleshooting information is fed back to the functional development stage to support debugging and redevelopment of the actual VNF code, thereby closing the DevOps life cycle loop across dimensions.

Conclusion

The stringent requirements of 5G networks are driving the need for further adaptation of existing DevOps practices and toolchains to the telecom industry. Our work with the open source and academic communities demonstrates how to address the 5G challenges related to the evolution of classic telecom fulfillment and assurance processes toward DevOps-powered cycles. Doing so requires an architecture that supports

automated deployment and operations, using powerful description languages tailored to different system dimensions that can capture constraints and feature specifications. Transparency of state changes and transitions throughout the architecture enables efficient operations. Our experience in the OPNFV community shows that CD practices including feedback loops throughout the technology stack and across organizations are key to a successful DevOps implementation. *

Further reading

- » **Open Daylight: The Journey to a DevOps Future (September 2016)**, available at: https://www.youtube.com/watch?v=llLxeD6Kwbs&index=10&list=PL8F5jrwEpGAiRCzJlyboA8Di3_TAjTT-2
- » **Achieving DevOps for NFV Continuous Delivery on Openstack (Verizon case study)**, available at: <https://www.openstack.org/videos/video/achieving-devops-for-nfv-continuous-delivery-on-openstack-verizon-case-study>
- » **State of DevOps Report 2016 (PuppetLabs)**, available at: <https://puppet.com/resources/whitepaper/2016-state-of-devops-report>
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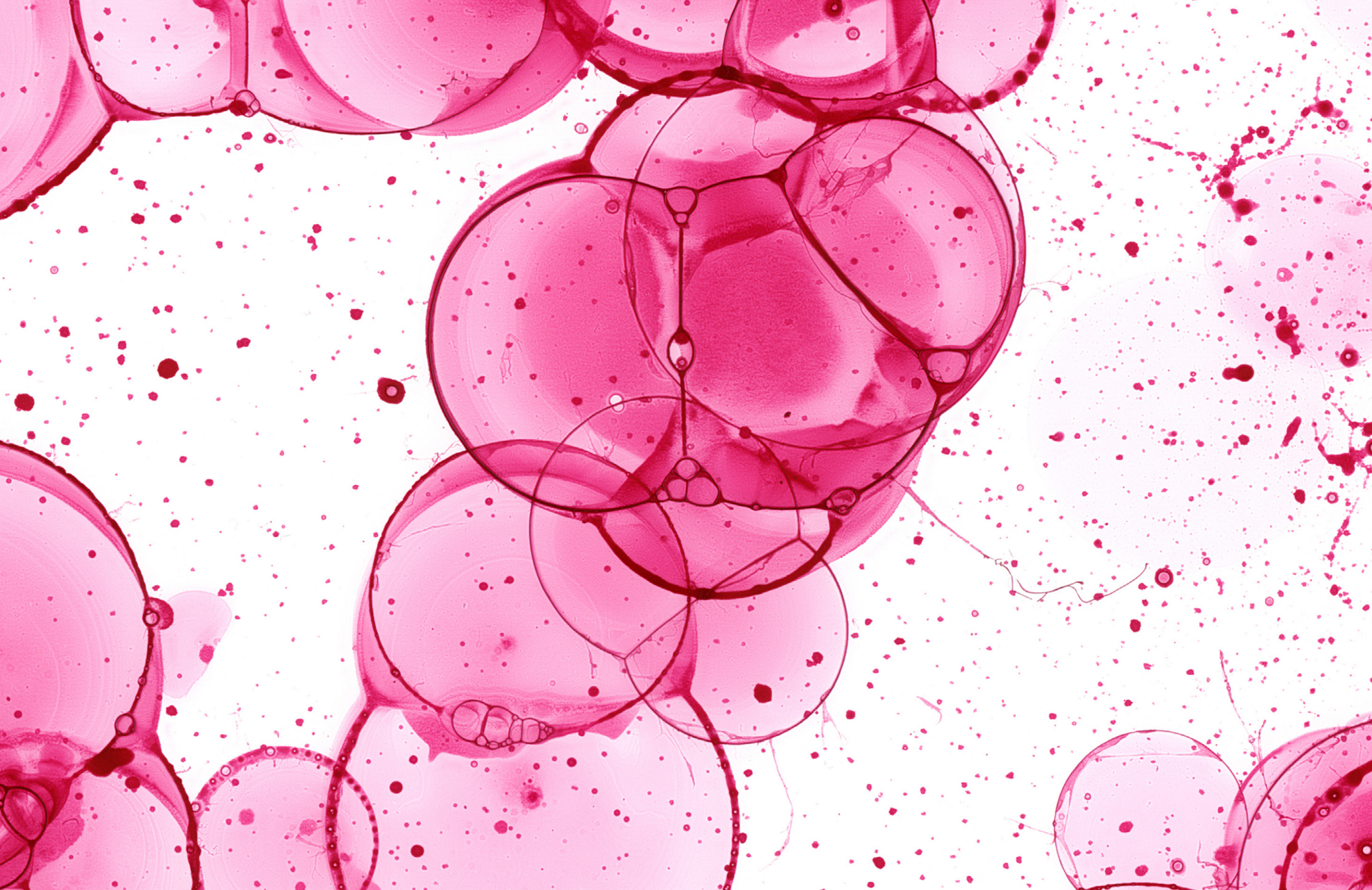
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